Study on Covariance Matrix Estimation

Part 2: Statistical and Economic Comparison on Equity Index Data

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Evaluation Framework

1 Objective and General Approach

The estimation of a covariance matrix inherently involves estimation uncertainty. In practice, the true covariance of returns is never observable from real data, which makes any direct evaluation of the error made by an estimator difficult. The objective of this work is therefore to compare different covariance estimation models in order to identify those that most faithfully approximate this unknown true matrix.

To this end, the evaluation is conducted along two complementary axes.

The first axis is statistical. It consists of placing the analysis in a controlled environment where the true covariance matrix is explicitly specified by construction. From this covariance, synthetic returns are generated, on which the different models are applied. This approach allows direct comparison of the estimated matrices to the true covariance and measurement of the estimation error in a perfectly controlled setting.

The second axis is economic. The estimators are then evaluated in a realistic portfolio management framework, where the objective is to minimize the tracking error of a portfolio relative to its reference index. The performance of the models is compared based on backtest results, in order to determine which estimators lead to the most efficient allocations in an operational context.

These two approaches serve different but complementary objectives. The statistical evaluation analyzes the intrinsic quality of an estimator independently of any operational consideration, while the economic evaluation measures its concrete impact on the performance of a real portfolio.

2 Study Universe and Data

The study covers the **MSDEEMUN** and **M1US** indices, comprising approximately 200 and 500 equities respectively. Price data are sourced from Bloomberg over the period from 02/01/2017 to 31/12/2025, representing approximately 2,270 trading days. Returns are computed as arithmetic first differences of prices.

The assets composing the index evolve over time. For the backtest exercises, at each rebalancing date, only the securities effectively present in the index are taken into account in the covariance and allocation computations, which excludes any survivorship bias. Absent assets are not included in the estimations, so that the effective dimension of the problem corresponds only to the actually investable universe at each date, without artificially inflating the size of the covariance matrix.

Benchmark weights correspond to the capitalization weights published by the index provider at quarterly frequency. Between two successive rebalancing dates, these weights are not held constant but evolve mechanically as a function of the returns of the underlying assets, then renormalized so that their sum equals one at each date. This dynamic reflects the natural drift of weightings induced by price fluctuations, consistent with the standard functioning of capitalization-weighted indices.

The performance of the index reconstructed in this study may differ marginally from those published by data providers, insofar as adjustments related to dividends and corporate actions are not explicitly taken into account. The prices used, extracted from standard data sources (Bloomberg), have not been subject to specific processing to integrate all idiosyncratic events affecting each company. This approximation does not, however, affect the conclusions of the study, as all models are evaluated consistently on this same price basis, which constitutes the reference to replicate in the economic backtests.

3 Models Evaluated

Five estimators are compared in this study. Each of them is presented and analyzed in detail in a study dedicated to the description and understanding of the different covariance estimation models.

Rolling Empirical. The sample covariance computed over a rolling window of w days constitutes the reference estimator. It is non-parametric, requires no structural assumption, and serves as a comparison point for all other models.

EWMA. Exponential smoothing according to the RiskMetrics formulation computes the covariance by recursion. Unlike the rolling window, all past observations contribute with exponentially decaying weights, which gives this estimator responsiveness to regime changes without abrupt cutoffs.

Ledoit-Wolf linear shrinkage (2004). The estimator combines the sample covariance and a spherical target with an optimal intensity, estimated directly from the data and requiring no external parameter.

Analytical nonlinear shrinkage (ANLS, 2020). Each eigenvalue of the sample matrix S is individually corrected by exploiting the local spectral density estimated by the Epanechnikov kernel.

Quadratic-Inverse Shrinkage (QIS, 2022). By optimizing the error on the precision matrix (the inverse of the covariance matrix) rather than on the covariance itself, QIS computes the corrected eigenvalues via a quadratic formula in the inverse space, which more precisely corrects the small eigenvalues whose inversion amplifies errors quadratically.

For each model, several lookback windows are tested, expressed in multiples of the annualization factor: a one-year window ($w = 252$ days or 52 weeks), two years, and three years. For the EWMA model, the decay parameter is set to $\lambda = 0.94$, consistent with the standard value proposed by RiskMetrics for daily returns.

Table 1: Correspondence between acronyms and full model names

Acronym	Full name	Reference
Rolling	Sample covariance over rolling window	–
EWMA	Exponentially Weighted Moving Average (RiskMetrics)	RiskMetrics, 1996
LW	Ledoit-Wolf linear shrinkage	Ledoit & Wolf, 2004
ANLS	Analytical nonlinear shrinkage	Ledoit & Wolf, 2020
QIS	Quadratic-Inverse Shrinkage	Ledoit & Wolf, 2022

Statistical Evaluation

1 Simulation Protocol

1.1 General Principle

The statistical evaluation rests on a controlled procedure in which synthetic returns are generated from a perfectly known true covariance. Each estimation model receives these simulated returns, produces an estimate of the covariance matrix, and the discrepancy between the estimated covariance and the true covariance is then measured using several loss metrics.

The true covariance Σ is entirely generated synthetically and is never estimated from real data. The dimensions of the problem, in particular the number of assets p and the sample length T, are set directly by the simulation protocol. No empirical calibration is used, with the possible exception of the target daily volatility level, imposed ex ante to ensure a realistic scale.

In order to assess the robustness of the estimators against the variability induced by randomness, the experiment is repeated over several independent scenarios, obtained from distinct random seeds. The reported statistics correspond to the means of the loss metrics over this set of scenarios, providing a stable comparison base across the different models with as little asymptotic bias as possible.

1.2 Two Data Generating Processes

Two data generating processes are implemented, each addressing a different question.

1.2.1 DGP Static Oracle

The first process, called *Static Oracle*, generates a single true covariance that is fixed throughout the scenario. The procedure decomposes into three successive steps.

Step 1 — Drawing the true covariance $\boldsymbol{\Sigma}$. A covariance matrix with factor structure is constructed:

$$\boldsymbol{\Sigma} = \underbrace{BB^\top}_{\text{factor component}} + \underbrace{\operatorname{diag}(\sigma^2_{\text{idio}})}_{\text{idiosyncratic component}} \in \mathbb{R}^{p\times p}$$

where $B \in \mathbb{R}^{p\times K}$ is a matrix of i.i.d. Gaussian loadings and $\sigma^2_{\text{idio}} \sim \text{LogNormal}(0, 0.5)$ is a vector of idiosyncratic variances. The matrix is rescaled so that the target average daily variance is $\bar{\sigma}^2 = (1\%)^2$. $\boldsymbol{\Sigma}$ is drawn once per scenario. It represents the unobservable truth that each estimator seeks to approximate.

Step 2 — Simulating T i.i.d. observations. Conditionally on the true covariance matrix Σ, returns are generated as independent observations from a multivariate Gaussian distribution.

More precisely, for $t = 1, \ldots, T$:

$$z_t \sim \mathcal{N}(0, I_p), \qquad r_t = \mu + L_\Sigma z_t, \qquad L_\Sigma = \mathrm{chol}(\Sigma),$$

where z_t is a standard Gaussian vector, L_Σ denotes the Cholesky factor of Σ (i.e. $\Sigma = L_\Sigma L_\Sigma^\top$), and μ is an optional drift vector. In the statistical analysis, $\mu = 0$ is imposed in order to focus exclusively on estimating the covariance structure, without introducing a mean effect. By construction, this transformation guarantees that the simulated returns follow a zero-mean normal distribution with variance defined by the true covariance matrix:

$$r_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \Sigma).$$

The number of observations T is governed by the ratio $c = p/T$ (with p the number of assets):

c	T	Regime	Interpretation
1.0	$T = p$	Hard	S is singular
0.5	$T = 2p$	Intermediate	Significant spectral bias
0.2	$T = 5p$	Easy	S well-conditioned, reduced gaps

Step 3 — Estimation and loss measurement. Each estimator receives the T simulated returns and produces a single estimated covariance matrix $\hat{\boldsymbol{\Sigma}}$. The discrepancy between $\hat{\boldsymbol{\Sigma}}$ and the true covariance $\boldsymbol{\Sigma}$ is then evaluated using several loss metrics, described in detail in Section 2. Each metric provides a scalar value per scenario, enabling direct comparison of estimator performance.

Static Oracle protocol summary.

Action	Result
Draw B, σ^2_{idio}	$\boldsymbol{\Sigma} = BB^\top + \mathrm{diag}(\sigma^2_{\text{idio}})$ fixed
Simulate $r_1, \ldots, r_T \sim \mathcal{N}(0, \boldsymbol{\Sigma})$	Single path of T i.i.d. obs
Estimate $\hat{\boldsymbol{\Sigma}}$ (Rolling / LW / ANLS / QIS / EWMA)	One matrix $\hat{\boldsymbol{\Sigma}}$ per estimator
Compute $\|\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_F$, $\mathcal{L}_{\text{Stein}}(\hat{\boldsymbol{\Sigma}}, \boldsymbol{\Sigma})$, ...	Comparable scalar losses
Repeat over K independent scenarios	Distribution of losses

1.2.2 DGP Factor Shock

The second process, called *Factor Shock*, generates a true covariance $\boldsymbol{\Sigma}_t$ that varies at each date according to an AR(1) dynamic applied separately to the factor loadings and the log-idiosyncratic variances.

Step 1 — Initialization. The initial states are drawn:

$$B_0 \sim \mathcal{N}\left(0, \frac{1}{K}\right)^{p \times K}, \qquad \ell d_0 \sim \mathcal{N}(0, 1)^p$$

Drawing B_0 initializes the factor loadings with a normalized scale $1/\sqrt{K}$ in order to control the amplitude of factor risk as the number of factors increases.

Drawing ℓd_0 initializes the log-idiosyncratic variance levels around a unit mean value, providing a neutral starting point for the persistent dynamics of idiosyncratic risks.

Step 2 — AR(1) dynamics. At each date $t = 1, \ldots, T$, the true covariance matrix is updated via two stationary AR(1) processes:

$$B_t = \rho_B \, B_{t-1} + \underbrace{\sqrt{1-\rho_B^2}\,\sigma_B}_{\text{shock standard deviation}} \quad \varepsilon_t^B, \qquad \varepsilon_t^B \sim \mathcal{N}(0,1)$$

$$\ell d_t = \rho_d \, \ell d_{t-1} + \underbrace{\sqrt{1-\rho_d^2}\,\sigma_d}_{\text{shock standard deviation}} \quad \varepsilon_t^d, \qquad \varepsilon_t^d \sim \mathcal{N}(0,1)$$

The true covariance at date t is then:

$$\mathbf{\Sigma}_t = \underbrace{B_t B_t^\top}_{\text{factor risk}} + \underbrace{\operatorname{diag}(\exp(\ell d_t))}_{\text{idiosyncratic risk}}$$

$\mathbf{\Sigma}_t$ is strictly positive definite by construction. A global rescaling is applied at the end of the simulation so that the average daily variance is of the order of $(1\%)^2$.

Step 3 — Return simulation. At each date t, a return vector consistent with the instantaneous covariance is simulated:

$$r_t = L_t \, z_t, \qquad z_t \sim \mathcal{N}(0, I_p), \qquad L_t = \operatorname{chol}(\mathbf{\Sigma}_t)$$

The Cholesky decomposition transforms a standard Gaussian vector into correlated returns having exactly the covariance $\mathbf{\Sigma}_t$.

Step 4 — Dynamic loss evaluation. Each estimator receives the complete return path $\{r_1, \ldots, r_T\}$ and produces a sequence of conditional estimates $\{\hat{\mathbf{\Sigma}}_t\}_{t=1}^T$ of the covariance matrix. The estimation error is computed at each date using the loss metrics defined in Section 2, then aggregated by temporal average over the evaluation period.

Formally, for a given loss metric $\mathcal{L}$, the loss associated with a scenario and a model is defined by:

$$\overline{\mathcal{L}} = \frac{1}{T-w} \sum_{t=w+1}^{T} \mathcal{L}(\hat{\mathbf{\Sigma}}_t, \mathbf{\Sigma}_t),$$

where w denotes the lookback window of the estimator (e.g. 252 or 504 observations), and $\mathbf{\Sigma}_t$ the true covariance at date t. This aggregation produces a scalar value per scenario and per model, enabling direct comparison of the dynamic performance of the different estimators.

Factor Shock protocol summary.

Action	Result
Initialize B_0, ℓd_0	Initial states
For $t = 1 \to T$: AR(1) on B_t, ℓd_t	Path of T true covariances $\{\mathbf{\Sigma}_t\}$
For $t = 1 \to T$: $r_t = \mathrm{chol}(\mathbf{\Sigma}_t)\, z_t$	Path of T returns
Estimate $\{\hat{\mathbf{\Sigma}}_t\}$ (Rolling / LW / ANLS / QIS / EWMA)	One estimation path per model
Compute $\frac{1}{T-w}\sum_t \|\hat{\mathbf{\Sigma}}_t - \mathbf{\Sigma}_t\|_F$	Mean temporal loss
Repeat over K seed scenarios	Distribution of losses

This process tests the ability of estimators to adapt to a persistent covariance. In summary, returns remain conditionally non-autocorrelated, but their covariance matrix evolves over time as a function of the parameters, which constitutes a more realistic and demanding framework for the evaluation of covariance estimators.

2 Statistical Evaluation Metrics

Three loss metrics are computed for each pair $(\hat{\mathbf{\Sigma}}, \mathbf{\Sigma})$.

Frobenius norm. It measures the total squared error across all entries of the matrix:

$$\mathcal{L}_F(\hat{\mathbf{\Sigma}}, \mathbf{\Sigma}) = \|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_F = \sqrt{\sum_{i,j}(\hat{\sigma}_{ij} - \sigma_{ij})^2}.$$

This metric treats all entries symmetrically, whether variances or covariances. A high Frobenius norm means that the estimator is systematically wrong on many asset pairs.

Spectral norm. The spectral norm measures the maximum error made by the estimator in a given risk direction. It corresponds to the largest eigenvalue of the error matrix $\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}$, and can be interpreted as the variance error on the most unfavorable portfolio:

$$\mathcal{L}_S(\hat{\mathbf{\Sigma}}, \mathbf{\Sigma}) = \|\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}\|_{\mathrm{op}} = \lambda_{\max}(\hat{\mathbf{\Sigma}} - \mathbf{\Sigma}).$$

Unlike the Frobenius norm, which aggregates all errors globally, the spectral norm is particularly sensitive to errors concentrated on a few dominant directions, typically associated with the largest eigenvalues of the covariance. This property makes it especially relevant in a portfolio optimization context, where poor estimation of the large spectral components leads to significant biases in risks and allocations.

Stein loss. The Stein loss corresponds, up to a constant, to the Kullback–Leibler divergence between two normal distributions with covariances $\mathbf{\Sigma}$ and $\hat{\mathbf{\Sigma}}$. It is invariant under linear transformations of the data and writes:

$$\mathcal{L}_{\mathrm{Stein}}(\hat{\mathbf{\Sigma}}, \mathbf{\Sigma}) = \underbrace{\mathrm{tr}\Big(\hat{\mathbf{\Sigma}}\,\mathbf{\Sigma}^{-1}\Big)}_{\substack{\text{mean variance ratio error}\\ \text{sensitivity to relative errors}}} - \underbrace{\log\det\Big(\hat{\mathbf{\Sigma}}\,\mathbf{\Sigma}^{-1}\Big)}_{\substack{\text{volume distortion penalty}\\ \text{barrier against misestimated eigenvalues}}} - \underbrace{p}_{\substack{\text{normalization term}\\ \mathcal{L}_{\mathrm{Stein}}(\mathbf{\Sigma},\mathbf{\Sigma})=0}}.$$

In the eigenbasis of $\mathbf{\Sigma}$, the Stein loss writes as a sum of spectral penalties:

$$\mathcal{L}_{\text{Stein}} = \sum_{i=1}^{p} \left(\frac{\hat{\lambda}_i}{\lambda_i} - \log\left(\frac{\hat{\lambda}_i}{\lambda_i}\right) - 1 \right),$$

showing that each eigenvalue is penalized individually according to its relative error.

This metric penalizes relative errors across the entire spectrum of the covariance matrix, by explicitly comparing the ratios between estimated and true eigenvalues. Unlike the Frobenius norm, which gives uniform weight to entry-by-entry errors, the Stein loss is particularly sensitive to poor estimation of small eigenvalues. The latter play a critical role in portfolio optimization problems, where they control the risks associated with strategies using optimizers.

For each scenario k, the three metrics are computed. The final reported statistic is the mean $\bar{\mathcal{L}} = \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_k$.

3 Results: DGP Static Oracle

	$p/T = 0.2$ (easy regime)				$p/T = 1.0$ (hard regime)			
Model	$N_{\text{scen.}}$	Frobenius	Spectral	Stein	$N_{\text{scen.}}$	Frobenius	Spectral	Stein
ANLS	20 000	0.0071	0.0036	107.23	20 000	0.0061	0.0047	$\gg 10^6$
EWMA	20 000	0.0065	0.0049	887.84	20 000	0.0061	0.0047	300.68
LW	20 000	0.0024	0.0014	158.48	20 000	0.0029	0.0019	725.73
QIS	20 000	0.0025	0.0014	**41.56**	20 000	0.0031	0.0020	**87.01**
Rolling	20 000	0.0034	0.0022	160.48	20 000	0.0036	0.0024	249.39

Before interpreting the model hierarchies, it is useful to put the absolute values of the Frobenius norm in context. Recall that the target daily volatility is fixed at 1% ($\sigma = 0.01$), so that the diagonal variances of the true matrix Σ are of the order of $(0.01)^2 = 10^{-4}$, and the off-diagonal covariances are of a comparable order of magnitude. A Frobenius norm of 0.0024 for LW in the easy regime means that the total squared error across all $p \times p$ entries of the matrix is of that order, corresponding to an average entry-level error of the order of $0.0024/p$, very small relative to the covariance values themselves. These levels indicate that shrinkage estimators produce covariance matrices very close to the true structure entry by entry, which is relevant for risk budgeting applications where each covariance term between asset pairs is directly used. Conversely, a Frobenius norm of 0.0071 for ANLS, approximately three times higher, reflects notably larger estimation errors on individual covariances, with a potential impact on risk allocation across assets.

Several findings emerge from this aggregated table. On the Frobenius and Spectral metrics, LW and QIS dominate in the easy regime ($p/T = 0.2$), with losses 50 to 60% lower than those of Rolling and EWMA. In the hard regime ($p/T = 1.0$), the gaps narrow slightly on these two metrics, with all models converging toward similar levels. The Stein loss reveals a much sharper hierarchy: QIS achieves the lowest loss in both regimes (41.56 at $p/T = 0.2$ and 87.01

at $p/T = 1.0$), consistent with its theoretical optimality objective under this metric.

The Stein losses of ANLS and EWMA at $p/T = 1.0$ reach numerically unstable values (of the order of 10^{11} to 10^{12}). This phenomenon is explained by the fact that at $p/T = 1$, the sample matrix is on the boundary of singularity ($T = p$): its smallest eigenvalues tend to zero, causing the terms $\log\det(\hat{\Sigma}\Sigma^{-1})$ and $\mathrm{tr}(\hat{\Sigma}\Sigma^{-1})$ of the Stein loss to explode. Estimators that do not sufficiently regularize the spectrum are most exposed to this numerical instability.

3.1 Results by Innovation Type

Table 2: Mean losses by model and p/T ratio — Static Oracle, Gaussian innovation

	$p/T = 0.2$				$p/T = 1.0$			
Model	$N_{\text{scen.}}$	Frobenius	Spectral	Stein	$N_{\text{scen.}}$	Frobenius	Spectral	Stein
ANLS	10 000	0.0037	0.0018	44.74	10 000	0.0035	0.0023	$\gg 10^9$
EWMA	10 000	0.0036	0.0023	842.75	10 000	0.0035	0.0023	261.48
LW	10 000	0.0007	0.0004	26.51	10 000	0.0014	0.0008	200.98
QIS	10 000	0.0006	0.0003	**13.63**	10 000	0.0012	0.0007	**42.22**
Rolling	10 000	0.0013	0.0008	119.49	10 000	0.0015	0.0008	203.99

Table 3: Mean losses by model and p/T ratio — Static Oracle, Student innovation

	$p/T = 0.2$				$p/T = 1.0$			
Model	$N_{\text{scen.}}$	Frobenius	Spectral	Stein	$N_{\text{scen.}}$	Frobenius	Spectral	Stein
ANLS	10 000	0.0105	0.0053	169.72	10 000	0.0087	0.0071	$\gg 10^9$
EWMA	10 000	0.0093	0.0075	932.92	10 000	0.0087	0.0071	339.88
LW	10 000	0.0042	0.0024	290.46	10 000	0.0043	0.0030	1250.47
QIS	10 000	0.0044	0.0024	**69.94**	10 000	0.0051	0.0033	**131.80**
Rolling	10 000	0.0054	0.0036	201.47	10 000	0.0057	0.0039	294.78

The analysis by innovation type brings several important nuances.

Under Gaussian innovation at $p/T = 0.2$, LW reaches 0.0007 and QIS 0.0006, corresponding to entry-level errors of the order of $0.0007/p$, reflecting near-perfect estimation of the covariance structure in this favorable regime. These values constitute a theoretical floor attainable by shrinkage estimators when the framework assumptions are perfectly met — Gaussian returns, fixed covariance, moderate p/T ratio — and are directly relevant for risk budgeting applications: the covariances between asset pairs estimated by LW or QIS are sufficiently close to the true values for marginal risk contributions to be reliable.

Under Student innovation, the Frobenius losses of LW and QIS are multiplied by a factor of 6 to 7 (0.0042 and 0.0044 at $p/T = 0.2$). This multiplicative factor may seem large, but the absolute values remain low: a Frobenius norm of 0.0042 remains of the same order of magnitude as the covariances themselves, which are of the order of $(0.01)^2 = 10^{-4}$. Shrinkage estimators therefore maintain satisfactory precision even under heavy tails, and the resulting risk budgets remain theoretically exploitable. The impact of Student innovations manifests more strongly on the Stein loss, which is sensitive to small eigenvalues, than on the entry-by-entry precision measured by Frobenius.

The Frobenius and Spectral losses are systematically higher under Student innovation than under Gaussian innovation, for all models and in both regimes. This result is expected: the heavy tails of the Student distribution introduce extreme observations that perturb the estimation of the covariance structure, increasing the variability of the estimators. On Frobenius, losses are approximately two to three times higher under Student than under Gaussian for LW and QIS, and by a similar factor for Rolling.

A striking result concerns LW under Student innovation at $p/T = 1.0$, whose Stein loss reaches 1250.47, a level significantly higher than that observed under Gaussian (200.98). The heavy tails of Student produce outliers that distort the small eigenvalues of the sample matrix, precisely those that the Stein loss penalizes most severely. QIS resists this degradation better (131.80 under Student versus 42.22 under Gaussian at $p/T = 1.0$).

Under Gaussian innovation, the theoretical hierarchy is respected in a near-perfect manner. QIS dominates on all metrics in both regimes, followed by LW. The gap between QIS and LW is particularly pronounced on the Stein loss: QIS achieves 13.63 versus 26.51 for LW at $p/T = 0.2$, and 42.22 versus 200.98 at $p/T = 1.0$. This growing gap with p/T empirically illustrates the analytical superiority of QIS in underdetermined regimes. Rolling sits in an intermediate position on Frobenius and Spectral, while EWMA presents a high Stein loss (842.75 at $p/T = 0.2$) due to its insensitivity to spectral correction.

Under Student innovation, the hierarchy partially changes. LW becomes less performant than QIS on the Stein loss, but moves ahead of Rolling on Frobenius and Spectral at $p/T = 0.2$. More remarkably, QIS maintains its dominance on the Stein loss in both regimes, confirming that its spectral correction mechanism remains effective even in the presence of heavy tails. EWMA remains the least performant on Stein regardless of regime, with very high losses (932.92 at $p/T = 0.2$).

Static Oracle summary. The Static Oracle DGP empirically demonstrates a hierarchy of estimators in the i.i.d. framework. QIS dominates on the Stein loss in all regimes and under both innovation types, with an advantage that amplifies as p/T increases. LW and QIS jointly dominate on Frobenius and Spectral under Gaussian innovation, while heavy-tailed (Student) innovations penalize LW more than QIS on the Stein loss. Rolling behaves correctly but is systematically dominated by the shrinkage approaches. EWMA is structurally unsuited.

4 Results: DGP Factor Shock

The Factor Shock DGP evaluates the ability of estimators to track a non-stationary dynamic covariance. Unlike the Static Oracle, each scenario generates a complete path of T true covariances $\{\Sigma_t\}$ and associated returns, on which each estimator produces a sequence of conditional estimates $\{\hat{\Sigma}_t\}$. Losses are then averaged over the evaluation period.

Each parameter combination (innovation $\times$ N_{sim}) is evaluated over 1 000 independent scenarios, each corresponding to a new draw of initial states B_0 and ℓd_0 and therefore a new path of true covariances. The number of covariance matrices effectively compared per scenario depends on the model window:

- Models with a 252-day window: $1\,000 \times 553 = 553\,000$ comparisons per configuration.
- Models with a 504-day window: $1\,000 \times 301 = 301\,000$ comparisons per configuration.

All scenarios are generated with the following parameters: number of factors $K = 10$, loading persistence $\rho_B = 0.95$, loading shock amplitude $\sigma_B = 0.05$, log-idiosyncratic variance persistence $\rho_d = 0.97$, and idiosyncratic shock amplitude $\sigma_d = 0.10$. These values produce a covariance that evolves slowly and persistently: the high value of ρ_B implies that the factor structure changes little from one period to the next (95% of the past structure is preserved), while $\sigma_B = 0.05$ keeps loading shocks at a moderate level. Similarly, $\rho_d = 0.97$ imparts high inertia to idiosyncratic risk levels. This calibration aims to reproduce a realistic financial market regime, in which the covariance structure is persistent but not constant, testing the ability of estimators to adapt gradually to regime changes without overreacting to noise.

4.1 Aggregated Results — All Innovation Types

Table 4: Mean losses by model — Factor Shock (all innovations, $N_{\text{sim}} \in \{100, 500\}$)

Model	$N_{\text{scen.}}$	Frobenius	Spectral	Stein
ANLS_252	4 000	0.0048	0.0020	269.49
ANLS_504	4 000	27.8587	3.1996	4 547 727.42
EWMA_252	4 000	0.0086	0.0050	2348.23
EWMA_504	4 000	0.0087	0.0051	2042.83
LW_252	4 000	0.0007	0.0001	**28.66**
LW_504	4 000	0.0007	0.0001	28.60
QIS_252	4 000	0.0017	0.0010	56.79
QIS_504	4 000	0.0014	0.0007	52.81
Rolling_252	4 000	0.0034	0.0014	1764.01
Rolling_504	4 000	0.0025	0.0009	316.97

The first striking observation is the poor performance of ANLS_504 across all metrics (Frobenius = 27.86, Stein $\approx 4.5 \times 10^6$), reflecting a structural numerical instability of this estimator when the 504-day window is applied to a large-dimensional dynamic path. This result is driven by the $N_{\text{sim}} = 500$ scenarios, where the combination of high dimension and long window leads to spectrally distorted estimates.

Aside from this pathological case, the absolute Frobenius levels merit comment before examining the hierarchies. LW displays a Frobenius norm of 0.0007 on both windows, a remarkably low level which, relative to true covariances of the order of $(0.01)^2 = 10^{-4}$, indicates that LW manages to estimate the dynamic covariance structure with very satisfactory entry-level precision, including in a non-stationary framework. QIS presents slightly higher levels (0.0014 to 0.0017), but these remain low in absolute terms. EWMA and Rolling, with Frobenius values around 0.0025 to 0.0086, deviate more from the true matrix entry by entry, which would translate in practice into less precise risk budgets. These absolute gaps, though modest for LW and QIS, are significant relative to the orders of magnitude of the covariances themselves and fully justify the use of shrinkage estimators in risk budgeting applications.

LW clearly dominates across all metrics, with Frobenius and Spectral losses an order of magnitude lower than all other models (Frobenius = 0.0007, Spectral = 0.0001), and this for both windows. On the Stein loss, LW_504 achieves the best score (28.60) ahead of LW_252 (28.66), suggesting that window lengthening benefits LW slightly in this persistently dynamic framework. QIS follows in second place on all metrics (Stein = 56.79 for QIS_252, 52.81 for QIS_504). EWMA and Rolling present very high Stein losses (2348 and 1764 respectively for the 252-day windows), reflecting their structural difficulty in estimating the spectrum of a non-stationary covariance, particularly at large dimension.

4.2 Results by Innovation Type

Table 5: Mean losses by model and N_{sim} — Factor Shock, Gaussian innovation

	$N_{\text{sim}} = 100$				$N_{\text{sim}} = 500$			
Model	$N_{\text{scen.}}$	Frobenius	Spectral	Stein	$N_{\text{scen.}}$	Frobenius	Spectral	Stein
ANLS_252	1 000	0.0024	0.0006	99.54	1 000	0.0029	0.0005	216.14
ANLS_504	1 000	0.0009	0.0002	23.55	1 000	45.7655	5.2532	7 705 384.29
EWMA_252	1 000	0.0017	0.0007	200.46	1 000	0.0086	0.0032	4341.29
EWMA_504	1 000	0.0017	0.0007	199.25	1 000	0.0086	0.0032	3749.99
LW_252	1 000	0.0001	0.0000	0.72	1 000	0.0004	0.0001	5.13
LW_504	1 000	0.0001	0.0000	0.72	1 000	0.0004	0.0001	5.11
QIS_252	1 000	0.0001	0.0000	1.10	1 000	0.0005	0.0001	6.64
QIS_504	1 000	0.0001	0.0000	0.91	1 000	0.0005	0.0001	12.98
Rolling_252	1 000	0.0006	0.0002	24.22	1 000	0.0031	0.0005	3343.98
Rolling_504	1 000	0.0005	0.0001	11.49	1 000	0.0022	0.0003	492.28

Table 6: Mean losses by model and N_{sim} — Factor Shock, Student innovation

	$N_{\text{sim}} = 100$				$N_{\text{sim}} = 500$			
Model	$N_{\text{scen.}}$	Frobenius	Spectral	Stein	$N_{\text{scen.}}$	Frobenius	Spectral	Stein
ANLS_252	1 000	0.0057	0.0027	249.47	1 000	0.0083	0.0042	512.79
ANLS_504	1 000	0.0024	0.0010	87.17	1 000	65.6660	7.5439	10 485 414.67
EWMA_252	1 000	0.0041	0.0028	242.10	1 000	0.0201	0.0134	4609.07
EWMA_504	1 000	0.0041	0.0028	240.14	1 000	0.0203	0.0136	3981.94
LW_252	1 000	0.0007	0.0001	17.41	1 000	0.0015	0.0001	91.64
LW_504	1 000	0.0007	0.0001	17.39	1 000	0.0015	0.0001	91.19
QIS_252	1 000	0.0012	0.0007	26.30	1 000	0.0051	0.0033	193.12
QIS_504	1 000	0.0010	0.0005	23.29	1 000	0.0039	0.0023	174.26
Rolling_252	1 000	0.0018	0.0009	56.47	1 000	0.0082	0.0041	3631.81
Rolling_504	1 000	0.0013	0.0006	37.61	1 000	0.0060	0.0028	726.48

Under Gaussian innovation and $N_{\text{sim}} = 100$, LW and QIS display near-zero Frobenius and Spectral losses (0.0001), very close to each other. On the Stein loss, LW achieves 0.72 versus 1.10 for QIS. A notable reversal compared to the Static Oracle where QIS dominated. This result is explained by the nature of the DGP: in a dynamic framework with $\rho_B = 0.95$ (strong persistence), the true covariance evolves slowly following a linear process (AR(1)), which favors linear shrinkage.

The transition from $N_{\text{sim}} = 100$ to $N_{\text{sim}} = 500$ strongly degrades the performance of all non-shrinkage models. Rolling_252 sees its Stein loss explode from 24.22 to 3343.98 under Gaussian innovation. EWMA suffers a similar degradation (200 to 4341). LW resists much better: its Stein loss only increases from 0.72 to 5.13. QIS degrades more than LW on Stein (from 1.10 to 6.64 for QIS_252 versus 0.72 to 5.13 for LW_252), confirming that in this Gaussian dynamic framework LW retains a slight advantage. ANLS_504 at $N_{\text{sim}} = 500$ produces aberrant values ($\approx 7.7 \times 10^6$ on Stein), reflecting numerical instability in this high-dimensional dynamic regime.

The switch to Student innovation uniformly degrades all models, but unequally. LW remains the most robust model on Frobenius and Spectral in both dimensional regimes. On Stein at $N_{\text{sim}} = 500$, LW moves from 5.13 (Gaussian) to 91.64 (Student), versus 6.64 to 193.12 for QIS. Rolling suffers the most severe degradation: from 3343.98 to 3631.81 at $N_{\text{sim}} = 500$, a more moderate proportional increase but at an already very high absolute level. ANLS_504 remains unstable at large dimension.

Factor Shock summary. The Factor Shock DGP slightly modifies the hierarchy observed in the Static Oracle. LW takes the advantage over QIS, particularly on the Stein loss under Gaussian innovation, where its linear shrinkage is theoretically optimal in a persistent covariance framework modeled via a linear process. QIS remains competitive but degrades relatively more as dimension increases. Non-shrinkage models (Rolling, EWMA) structurally suffer from non-stationarity combined with large dimension: their Stein loss explodes with N_{sim}, confirming their direct exposure to the effective p/n ratio. ANLS with a long window (504 days) presents severe numerical instability at large dynamic dimension, making it an estimator to avoid in this regime. Finally, lengthening the window from 252 to 504 days benefits the stable models (LW, Rolling) but aggravates ANLS instability, illustrating that the window choice interacts strongly with the regularization mechanism of each estimator.

5 General Summary and Lessons from the Statistical Evaluation

The statistical evaluation conducted on the two DGPs allows us to draw structural lessons on the behavior of covariance estimators in controlled settings where the true matrix is perfectly known.

1. QIS dominates in the static i.i.d. framework. The Static Oracle DGP confirms the superiority of QIS on the Stein loss in all regimes and under both innovation types. This advantage is particularly clear at $p/T = 1.0$, where QIS achieves a Stein loss of 87.01 versus 200.98 for LW under Gaussian, and 131.80 versus 1250.47 under Student. This result empirically validates the theoretical objective of QIS: analytically correcting the spectral bias of the sample matrix in an optimal way under the Stein loss, including in singular regimes where $T = p$. On the Frobenius and Spectral metrics, LW and QIS are jointly in the lead, with losses two to three times lower than those of Rolling and EWMA.

2. LW takes the advantage in the dynamic framework. The Factor Shock DGP partially reverses this hierarchy. Under Gaussian innovation and low dimensionality ($N_{\text{sim}} = 100$), LW achieves a Stein loss of 0.72 versus 1.10 for QIS. This reversal is explained by the strong persistence of the DGP ($\rho_B = 0.95$): when the true covariance evolves slowly, the linear shrinkage of LW, designed for an i.i.d. framework but applied here on a rolling window, benefits from the temporal stability of the structure. QIS, whose spectral correction is more aggressive, slightly over-regularizes in this context. LW and QIS nonetheless jointly dominate all other models across all metrics and configurations.

3. Rolling and EWMA are structurally exposed to the p/n ratio. Both non-shrinkage models suffer severe and monotone degradation with dimensionality. On the Factor Shock, Rolling_252 sees its Stein loss increase from 24.22 at $N_{\text{sim}} = 100$ to 3343.98 at $N_{\text{sim}} = 500$ under Gaussian innovation. EWMA follows a similar trajectory. This direct exposure to the effective p/n ratio confirms that without a regularization mechanism, these estimators are unsuited to high-dimensional contexts, precisely those that characterize practical index replication applications on large universes.

4. EWMA unsuited to the i.i.d. framework, unstable at large dynamic dimension. EWMA presents the least favorable profile in both DGPs. In the Static Oracle, its exponential weighting mechanism, designed to track a varying covariance, is poorly adapted to a fixed covariance, producing systematically high Stein losses (842.75 at $p/T = 0.2$ under Gaussian). In the Factor Shock, it fails to exploit the persistence of the DGP, with Stein losses exploding at large dimension (4341 at $N_{\text{sim}} = 500$). EWMA thus appears as the least robust model across all tested configurations.

5. ANLS exhibits severe numerical instability at large dynamic dimension. ANLS_504 produces aberrant values ($\approx 10^6$ on Stein) in the Factor Shock scenarios at $N_{\text{sim}} = 500$, regardless of the innovation type. This pathological behavior, absent at low dimension, reveals a structural limitation of the ANLS estimator when applied with a long window in a high-dimensional dynamic context. ANLS_252 remains globally competitive, but is still dominated by LW and QIS on the Stein loss in both DGPs.

6. Student innovation penalizes all models, but QIS resists better. The heavy tails of the Student distribution distort the small eigenvalues of the estimated matrix, resulting in more pronounced Stein loss degradation for all models. QIS resists this perturbation better than LW: in the Static Oracle at $p/T = 1.0$, QIS maintains a Stein loss of 131.80 versus 1250.47 for LW under Student, confirming that its analytical spectral correction is more robust to heavy-tailed distributions.

7. Absolute error levels compatible with risk budgeting applications. Beyond the relative hierarchies between models, the absolute levels of the Frobenius norm provide a reassuring lesson for practical applications. Recall that the target daily volatility is fixed at 1%, so that the true covariances are of the order of $(0.01)^2 = 10^{-4}$. In this context, the Frobenius norms obtained by LW and QIS, of the order of 0.0007 to 0.0024 depending on the regime and DGP, remain very small relative to the orders of magnitude of the covariances themselves. These levels mean that, entry by entry, shrinkage estimators produce matrices very close to the true covariance structure, which is directly relevant for risk budgeting applications where each covariance term between asset pairs conditions marginal risk contributions. Even under Student innovation, where Frobenius losses are multiplied by a factor of 6 to 7, the absolute values remain low (0.0042 to 0.0044 for LW and QIS at $p/T = 0.2$): risk budgets constructed from these estimates remain theoretically exploitable. This finding reinforces the recommendation to use shrinkage estimators, LW or QIS depending on the regime, in any application where the precision of the estimated covariance matrix directly conditions allocation or risk measurement decisions.

8. Perspectives for parametric extensions. The results presented rely on a fixed calibration of DGP parameters. Several extensions would be worth exploring to deepen these conclusions. On the Static Oracle, testing intermediate p/T ratios (notably $p/T = 0.5$) would better characterize the transition regime between QIS dominance and model convergence. On the Factor Shock, varying ρ_B between low values (structure changes rapidly, $\rho_B \approx 0.5$) and high values ($\rho_B \approx 0.99$) would test the bias-variance trade-off of estimators under more contrasting non-stationarity regimes. Similarly, testing lower degrees of freedom for the Student innovation ($\nu = 3$ instead of $\nu = 5$ used here) would strengthen the heavy tails and allow evaluation of estimator robustness under more extreme market conditions. Finally, an extension with intermediate N_{sim} values (200, 300 assets) would allow tracing the degradation curve of models with dimensionality and more precisely identify the threshold at which ANLS_504 instability appears.

Economic Evaluation

1 Backtest Framework

1.1 Replication Strategy

The economic evaluation is conducted within an index replication strategy based on ex-ante tracking error minimization. At each rebalancing date τ, the portfolio solves the quadratic program:

$$w_\tau^* = \arg\min_{w \in \mathcal{C}} (w - b_\tau)^\top \hat{\mathbf{\Sigma}}_\tau (w - b_\tau)$$

where b_τ is the vector of benchmark weights at date τ, $\hat{\mathbf{\Sigma}}_\tau$ is the covariance matrix estimated by the model under consideration, and $\mathcal{C}$ is the set of portfolio constraints, defined by the budget condition $\mathbf{1}^\top w = 1$ (weights sum to one), the non-negativity constraint $w \geq 0$ (no short positions), and restriction to the investable universe only.

The optimization is solved by the Clarabel solver, which treats the problem as a conic quadratic program and guarantees global optimality. The estimated covariance $\hat{\mathbf{\Sigma}}_\tau$ is constructed by the model on the data window available up to $\tau - 1$, so that the entire estimation and optimization procedure is causal.

1.2 Investable Universe

The study universes correspond to the indices considered in the analysis. For each index, the theoretical universe is defined as the set of assets composing it at a given date. In order to reflect the constraints encountered in practice by asset managers, a restriction is then applied to this universe to define the effective investable universe.

These restrictions take the form of exclusions, which aim to simulate common portfolio construction constraints, in particular those related to responsible investment policies or internal limitations.

In the optimization programs, the objective remains the replication of the complete index, while the investable universe used for portfolio construction and authorized in the optimizer is restricted by these exclusions. This asymmetry between the target universe and the investable universe generates a structural tracking error, which the optimizer seeks to minimize as much as possible from the information provided by the estimated covariance matrix, which is itself constructed on the full universe.

1.3 Backtest Parameters

Table 7: Parameters of the backtest procedure

Parameter	Value
Evaluation period	02/01/2017 – 31/12/2025
Rebalancing frequency	Quarterly (index rebalancing dates + 1 day)
Data	Daily
Lookback windows tested	252, 504 and 756 for each model
Optimization solver	Clarabel (conic QP)
Transaction costs	Zero (comparison of estimators with identical structure)

2 Economic Metrics

Ex-post tracking error. The realized tracking error over the period $[t_0, t_1]$ is defined as the standard deviation of daily active returns, annualized using the factor $\sqrt{252}$:

$$\mathrm{TE}_{\text{ex-post}} = \sqrt{252} \cdot \sigma\left(r_t^p - r_t^b\right),$$

where r_t^p denotes the portfolio return and r_t^b the benchmark return at date t.

Ex-ante tracking error. The ex-ante tracking error predicted by the model at rebalancing date τ is defined as the standard deviation of the active portfolio computed from the estimated covariance matrix, then annualized using the factor $\sqrt{252}$:

$$\mathrm{TE}_{\text{ex-ante}}(\tau) = \sqrt{(w_\tau - b_\tau)^\top \hat{\boldsymbol{\Sigma}}_\tau (w_\tau - b_\tau) \cdot 252}.$$

The systematic gap between $\mathrm{TE}_{\text{ex-ante}}$ and $\mathrm{TE}_{\text{ex-post}}$ measures the forecast bias of the covariance model: a model that underestimates the tracking error leads to more concentrated and riskier portfolios than anticipated.

Information ratio. The information ratio is defined as the ratio of the annualized active return to the ex-post tracking error:

$$\mathrm{IR} = \frac{\bar{\alpha} \cdot 252}{\mathrm{TE}_{\text{ex-post}}}$$

where $\bar{\alpha}$ is the daily mean of active returns.

3 Economic Results

The economic results are presented for two distinct universes, in order to evaluate the impact of exclusion constraints on the ability of the models to replicate their reference index. In all cases, the optimization objective is the replication of the complete index, while the investable universe may be restricted by exclusions.

3.1 MSDEEMUN Index — Full Universe Without Exclusions

This first case corresponds to an investable universe identical to the MSDEEMUN index, without exclusions. It constitutes the reference case, in which the tracking error arises exclusively from covariance matrix estimation errors and optimization constraints.

Table 8: Economic metrics — MSDEEMUN without exclusions (common period 2020–2025)

Model	Ex-post TE	Cumulative perf.
Rolling 252d	0.10%	0.14%
Rolling 504d	0.10%	0.14%
Rolling 756d	0.10%	0.20%
EWMA 252d	0.22%	0.18%
EWMA 504d	0.22%	0.17%
EWMA 756d	0.20%	-0.05%
LW 252d	0.09%	0.16%
LW 504d	0.09%	0.15%
LW 756d	0.10%	0.22%
ANLS 252d	0.10%	0.15%
ANLS 504d	0.10%	0.14%
ANLS 756d	0.10%	0.20%
QIS 252d	0.09%	0.17%
QIS 504d	0.09%	0.18%
QIS 756d	0.10%	0.22%

In the full MSDEEMUN universe, tracking error levels are low and very close across models, which naturally limits the ability to discriminate between them. At 252 days, Rolling presents a tracking error of 0.10%, while LW and QIS sit slightly below at 0.09%. Window lengthening does not significantly modify these levels — they remain between 0.09% and 0.10% regardless of the window — but tends to reduce the differences between models. The trajectories of the shrinkage models and Rolling progressively converge toward quasi-identical values at 756 days.

The EWMA model constitutes the structural exception in this configuration: with levels around 0.22% on the 252- and 504-day windows, it stands out clearly from all other models. The shrinkage models, LW and QIS in the lead, retain a slight level advantage on short windows, but this advantage remains moderate.

These results are explained by the very nature of the universe studied: in the absence of exclusions, all assets of the index are available and near-direct replication is theoretically attainable. The observed tracking error therefore stems essentially from estimation noise and optimization constraints, and not from a structural impossibility of replication. In this weakly differentiating setting, the role of the covariance estimator remains limited.

The detailed annual results by estimation window and the ex-post tracking error charts are presented in Appendix C.1.

3.2 MSDEEMUN Index — Restricted Universe With Exclusions

In this second case, the investable universe is obtained by applying an exclusion policy to the MSDEEMUN index. A fixed share of the constituents, representing on average *[3.60 %]* of the index capitalization, is excluded in order to simulate responsible investment constraints observed in practice.

Table 9: Economic metrics — MSDEEMUN with exclusions (period 2020–2025)

Model	Ex-post TE	Cumulative perf.
Rolling 252d	0.81%	-0.47%
Rolling 504d	0.58%	-0.27%
Rolling 756d	0.54%	-0.36%
EWMA 252d	0.95%	-0.88%
EWMA 504d	0.95%	-0.88%
EWMA 756d	0.92%	-1.50%
LW 252d	0.81%	-0.43%
LW 504d	0.53%	0.40%
LW 756d	0.52%	0.22%
ANLS 252d	0.81%	-0.43%
ANLS 504d	0.58%	-0.27%
ANLS 756d	0.54%	-0.36%
QIS 252d	0.58%	0.29%
QIS 504d	0.54%	0.50%
QIS 756d	0.52%	0.22%

The introduction of exclusions profoundly changes the level of tracking error and the ability to discriminate between models, placing the covariance estimator in a central role. On the short 252-day window, corresponding to a p/n ratio close to 1 for MSDEEMUN (approximately 240 assets), tracking error levels are highest and gaps between models most pronounced. Rolling, ANLS and LW all display a tracking error of 0.81%, while QIS stands out clearly at 0.58%, demonstrating its ability to stabilize estimation in a heavily noisy setting where the high p/n ratio penalizes less regularized estimators. EWMA remains structurally behind with levels between 0.92% and 0.95%, little sensitive to window lengthening, reflecting an inability to correct the structural gap introduced by the exclusions.

Window lengthening mechanically reduces the p/n ratio and progressively improves estimates for all models except EWMA. Rolling moves from 0.81% to 0.58% then 0.54%, directly illustrating the gain from reducing the p/n ratio. Shrinkage models also benefit from this lengthening but starting from an already lower level, as their regularization mechanism partially compensates for the bias related to the high p/n ratio on short windows. At 504 days, LW reaches 0.53%

and QIS 0.54%, versus 0.58% for Rolling and ANLS. At 756 days, LW and QIS converge to 0.52%, Rolling and ANLS to 0.54%, with EWMA remaining alone structurally above 0.92%.

Despite the overall improvement from window lengthening, the hierarchy between models remains stable and persistent. QIS establishes itself as the most robust model on the short window, where the p/n ratio is most unfavorable. LW and QIS jointly dominate on long horizons. Unlike the case without exclusions, the gaps between models are here structural. The impossibility of directly replicating the index reinforces the role of covariance estimation, and methodological choices translate directly into tracking error differences. Shrinkage approaches, particularly QIS and LW, thus appear more robust to estimation uncertainty linked to high dimension and offer persistently superior performance, regardless of window size.

The detailed annual results by estimation window and the ex-post tracking error charts are presented in Appendix C.2.

3.3 M1US Index — Full Universe Without Exclusions

This case corresponds to an investable universe identical to the M1US index, without exclusions. It constitutes the reference case, in which the tracking error arises exclusively from covariance matrix estimation errors and optimization constraints. The universe is however larger than MSDEEMUN (on average nearly 500 securities), which increases the high-dimensional character of the problem and reinforces the role of covariance estimation.

Table 10: Economic metrics — M1US without exclusions (period 2020–2025)

Model	Ex-post TE	Cumulative perf.
Rolling 252d	0.34%	0.55%
Rolling 504d	0.19%	0.37%
Rolling 756d	0.19%	0.28%
EWMA 252d	0.39%	1.07%
EWMA 504d	0.39%	1.09%
EWMA 756d	0.35%	0.13%
LW 252d	0.17%	0.49%
LW 504d	0.17%	0.53%
LW 756d	0.18%	0.28%
ANLS 252d	0.17%	0.49%
ANLS 504d	0.17%	0.54%
ANLS 756d	0.18%	0.27%
QIS 252d	0.17%	0.47%
QIS 504d	0.17%	0.50%
QIS 756d	0.18%	0.28%

In the full M1US universe, tracking error levels are noticeably higher than in the MSDEEMUN case, a direct reflection of the significantly larger universe (nearly 500 securities) and the more unfavorable p/n ratio. On the short 252-day window, this ratio is close to 2, which accentuates estimation noise and allows better discrimination of the different approaches than with MSDEEMUN. Rolling reaches a tracking error of 0.34%, approximately double that observed on the European index, while the shrinkage models (LW, ANLS, QIS) sit around 0.17%, approximately half of Rolling. This gap of nearly 17 basis points concretely illustrates the stability gain provided by shrinkage in a high-dimensional regime where $p/n \approx 2$. EWMA presents systematically higher levels (0.35%–0.39%), little sensitive to window lengthening, reflecting a persistent sensitivity to recent variations and a reduced ability to smooth estimation noise at large dimension.

Window lengthening mechanically reduces the p/n ratio and improves performance for all models, but in a differentiated manner. Rolling benefits the most from this lengthening: it moves from 0.34% at 252 days to 0.19% at 504 and 756 days, confirming its strong dependence on the p/n ratio. The shrinkage models, starting from an already lower level, converge toward 0.17% at 504 days then 0.18% at 756 days, retaining a structural advantage over Rolling that progressively attenuates. Differences between shrinkage models remain limited: LW and ANLS present the lowest levels on short and intermediate windows, with QIS displaying slightly superior stability during stress episodes, notably in 2022 and 2023 where the shrinkage models remain contained between 0.07% and 0.18% while Rolling and EWMA display significantly higher values.

On the long 756-day window, convergence between models is near-complete: ANLS, LW, QIS and Rolling all sit between 0.18% and 0.19%, with near-superimposed trajectories. Only EWMA remains behind at 0.35%, with persistent peaks confirming a structural sensitivity to recent variations that does not dissipate with window lengthening. With annualized active returns being very low for all models, the observed differences stem essentially from the tracking error level and not from a directional performance bias.

In summary, the M1US universe without exclusions reveals a stable hierarchy consistent with theoretical lessons: shrinkage models structurally dominate on short windows where the p/n ratio is most unfavorable, Rolling progressively approaches as n increases without ever equaling them on short windows, and EWMA remains structurally the least suited to this high-dimensional framework.

The detailed annual results by estimation window and the ex-post tracking error charts are presented in Appendix C.3.

3.4 M1US Index — Restricted Universe With Exclusions

This last case concerns the M1US index, for which a restricted investable universe is also considered. On average *[8.76 %]* of the index constituents are excluded.

Table 11: Economic metrics — M1US with exclusions (period 2020–2025)

Model	Ex-post TE	Cumulative perf.
Rolling 252d	0.87%	1.95%
Rolling 504d	0.54%	2.36%
Rolling 756d	0.49%	1.66%
EWMA 252d	0.77%	2.13%
EWMA 504d	0.78%	2.04%
EWMA 756d	0.71%	1.31%
LW 252d	0.53%	1.88%
LW 504d	0.49%	3.23%
LW 756d	0.44%	1.46%
ANLS 252d	0.58%	1.85%
ANLS 504d	0.63%	2.86%
ANLS 756d	0.57%	1.70%
QIS 252d	0.53%	1.98%
QIS 504d	0.50%	2.11%
QIS 756d	0.49%	1.66%

The introduction of exclusions in the M1US universe leads to a significant increase in tracking error levels and a much more pronounced differentiation between models than in the full universe. With a p/n ratio close to 2 on the short 252-day window, empirical estimation of the covariance matrix is particularly unstable. On this window, LW and QIS stand out at 0.53%, while ANLS sits at 0.58%, clearly behind the first two shrinkage estimators. EWMA follows (0.77%) and Rolling reaches 0.87%, the highest level, confirming its strong exposure to the high p/n ratio.

Window lengthening mechanically reduces the p/n ratio and progressively improves performance for all models. Rolling benefits the most from this reduction: it moves from 0.87% at 252 days to 0.54% at 504 days, then 0.49% at 756 days. LW stands out particularly on long windows: it reaches 0.49% at 504 days and 0.44% at 756 days, the best level among all models on this horizon, slightly ahead of QIS (0.49% at 756 days) and Rolling (0.49%). ANLS in contrast does not benefit from lengthening in the same way: its levels remain around 0.57%–0.63%, close to Rolling on intermediate and long windows, suggesting estimation noise sensitivity comparable to the empirical estimator in this large-dimensional dynamic regime. EWMA remains systematically behind with levels between 0.71% and 0.78%, little sensitive to window lengthening, reflecting a structural dependence on recent observations that does not dissipate with increasing n.

Unlike the case without exclusions, the gaps between models are here persistent and economically significant. The impossibility of directly replicating the index reinforces the role of covariance estimation. LW and QIS establish themselves as the most robust methods, with an advantage that becomes more pronounced on long windows. ANLS, despite belonging to the shrinkage family, approaches Rolling on intermediate and long horizons in this universe, which constitutes a notable result relative to the other tested configurations. EWMA remains structurally the least suited to this high-dimensional constrained framework.

The detailed annual results by estimation window and the ex-post tracking error charts are presented in Appendix C.4.

3.5 General Summary and Lessons from the Economic Backtest

The full set of results obtained on the two studied indices (MSDEEMUN and M1US), with and without exclusions, allows us to draw several structural lessons on the behavior of covariance estimation models in real conditions. Despite very different tracking error levels depending on the universe considered, the conclusions are remarkably consistent across indices and configurations.

1. The central role of the p/n ratio. One of the most robust results of this study is the determining impact of the p/n ratio on estimation stability. When the estimation window is short (252 days), the ratio is high — close to 1 in MSDEEMUN with exclusions, and close to 2 in M1US — which leads to higher tracking error levels and greater dispersion across models. As the window lengthens (504 then 756 days), the p/n ratio decreases mechanically, reducing estimation uncertainty and bringing the models closer together. This effect is particularly visible on the Rolling model, whose tracking error decreases strongly as n increases, confirming its sensitivity to statistical noise.

2. The structural superiority of shrinkage models. Beyond the mechanical effect of window lengthening, the results show that shrinkage models retain a clear and persistent advantage in all tested configurations. This advantage is visible even when the dimension is high and the p/n ratio is unfavorable. For example, on the short 252-day window on M1US with exclusions, LW and QIS reach 0.53%, versus 0.58% for ANLS, 0.87% for Rolling and 0.77% for EWMA. On MSDEEMUN without exclusions, where the tracking error is very low, QIS and LW remain systematically in the lead with levels around 0.09%, while Rolling and ANLS sit at 0.10%.

These results confirm that shrinkage models manage to stabilize covariance estimation even in unfavorable contexts, by reducing estimation noise without introducing excessive bias.

3. A stable hierarchy across models. The four studied configurations reveal a remarkably stable hierarchy across models. Across all configurations, shrinkage models dominate Rolling and EWMA, and Rolling structurally leads EWMA:

$$\{\text{QIS, LW}\} > \text{ANLS} > \text{Rolling} > \text{EWMA}.$$

QIS and LW jointly occupy the top position, with a slight alternation depending on the configuration: QIS leads on short windows and in the noisiest regimes (high p/n ratio), while LW takes the advantage on long windows, notably on M1US with exclusions where it reaches 0.44% at 756 days versus 0.49% for QIS. This alternation reflects the theoretical properties of each: QIS corrects the spectrum more aggressively in unfavorable regimes, LW benefits more from temporal stability when n is large. ANLS sits systematically behind the first two shrinkage estimators, sometimes approaching Rolling on long windows and large-dimensional universes. Rolling progressively approaches the shrinkage models as n increases, but never manages to equal them on short windows. EWMA remains structurally the least performant, with systematically higher tracking error levels and little sensitivity to window lengthening.

4. A first validation of the two central hypotheses of the study. The economic backtest results confirm the two key points this study sought to establish:

- **(i) Increasing the estimation window significantly reduces uncertainty**, in particular for noise-sensitive models such as Rolling.
- **(ii) Shrinkage models are globally superior**, with a stable ranking: QIS in the lead, followed by LW then ANLS.

These conclusions are robust to variations in index, dimension, exclusion structure and period.

5. Transition toward empirical validation. This first part relies on a single observed market trajectory. While it identifies strong trends, it is not sufficient to establish the robustness of the models across varied contexts. The following part extends this analysis by evaluating the models on a massive set of simulated scenarios, where the operational parameters (estimation window, rebalancing frequency, exclusion fraction, data frequency) are systematically varied. This validation assesses the stability of the conclusions obtained and determines whether the hierarchies observed in the real backtests are maintained in a broader framework less dependent on a particular trajectory.

Empirical Validation

1 Objective and Protocol

The economic evaluation presented in the previous part relies on a single observed market trajectory for each index. A single realization is not sufficient to establish robust conclusions. The realized tracking errors observed in the previous part depend on the specific market trajectory, the rebalancing dates, and the composition of the investable universe. Results favorable to a model may simply reflect a better fit with the particular structure of the tested period rather than an intrinsic superiority of the model. The validation proposed in this part addresses this limitation by simulating a large number of variants of the backtest experiment on our two universes. The true return data are preserved, but the parameters defining the application context are systematically varied.

Several dimensions of variability are explored in order to assess the robustness of covariance estimation models in different contexts. The first concerns the exclusion fraction applied to the reference index: for each exclusion rate $f \in \{5\%, 10\%, 20\%, 30\%, 40\%, 50\%\}$, several investable universes are generated using distinct random seeds, in order to simulate different plausible exclusion schemes encountered in practice.

The second dimension is the portfolio rebalancing frequency, tested at weekly, monthly, and quarterly frequencies. The third concerns the data window used by the covariance models for estimation, considered over different lengths corresponding to 0.5, 1, 1.5, 2, and 3 years. Finally, estimations are conducted on both daily and weekly data, in order to analyze the impact of observation frequency on model performance.

For each covariance estimation model, all combinations of these parameters are evaluated. Each configuration constitutes a distinct scenario, enabling systematic and homogeneous comparison of the models across a wide range of realistic situations.

The complete scenario grid is:

$$\begin{aligned}&\{\text{nb models}\} \times \{w \text{ (data window)}\}\\ &\times \{f \text{ (excluded fraction)}\} \times \{S \text{ (scenarios)}\} \times \{\text{rebalancing frequencies}\}\end{aligned}$$

In total, the evaluation covers all considered parameter combinations. For each covariance estimation model, 100 distinct investable universes are generated for each of the retained exclusion fractions, yielding 600 different investable universes per index. On each of these universes, backtests are conducted for several estimation windows, two data frequencies, and three rebalancing frequencies.

This parameter grid leads to a total of 18 000 independent scenarios per model, and 72 000 scenarios in total for all models evaluated on a single index. Each scenario corresponds to a specific configuration of the tracking error minimization problem.

The analysis does not only aim to compare models for a given configuration, but also allows evaluation of their robustness across a wide range of effective investable universes and operational parameters. In this sense, the results of this part extend and generalize those obtained in the first part of the backtests, placing them in a broader framework less dependent on a particular parameter choice.

2 Aggregated Results

The results presented in this section cover four models: Rolling, LW, ANLS and QIS. The EWMA model was excluded from this phase of analysis in order to reduce computation times, path-based estimates being particularly costly on the considered scenario grid. Furthermore, the results from the previous parts consistently showed that EWMA is structurally unsuited to the high-dimensional index replication framework, with systematically higher tracking error levels that are little sensitive to operational parameters. Its exclusion therefore does not deprive the analysis of discriminating information.

2.1 Distribution of Ex-Post Tracking Error

The following tables present, for each model and each index, the aggregated statistics of the annualized ex-post tracking error over all 18 000 evaluated scenarios.

Table 12: Aggregation by model — MSDEEMUN (mean over all scenarios)

Model	$N_{\text{scenarios}}$	Mean TE	Std TE	Min TE	Max TE
ANLS	18 000	1.45%	0.71%	0.17%	3.51%
LW	18 000	1.38%	0.71%	0.16%	3.57%
QIS	18 000	1.35%	0.70%	0.16%	3.57%
Rolling	18 000	1.81%	0.81%	0.19%	5.42%

Table 13: Aggregation by model — M1US (mean over all scenarios)

Model	$N_{\text{scenarios}}$	Mean TE	Std TE	Min TE	Max TE
ANLS	18 000	1.60%	0.89%	0.18%	4.83%
LW	18 000	1.57%	0.88%	0.18%	4.28%
QIS	18 000	1.53%	0.87%	0.18%	4.50%
Rolling	18 000	2.03%	0.92%	0.23%	4.98%

Across both universes, the model hierarchy is consistent with previous observations. The shrinkage approaches — QIS, LW and ANLS — display the lowest mean tracking errors. On MSDEEMUN, QIS achieves the best mean (1.35%), followed by LW (1.38%) and ANLS (1.45%).

On M1US, the same hierarchy is observed with QIS at 1.53%, LW at 1.57% and ANLS at 1.60%. The gaps between the three shrinkage models remain limited, of the order of 5 to 7 basis points on both universes.

Rolling presents noticeably higher levels, 1.81% on MSDEEMUN and 2.03% on M1US, respectively 46 and 50 basis points above QIS. Its dispersion is also the highest (0.81% on MSDEEMUN, 0.92% on M1US), confirming greater sensitivity to unfavorable configurations. With a maximum of 5.42% on MSDEEMUN, Rolling is particularly exposed to high-exclusion scenarios combined with short windows.

These results, aggregated over 18 000 scenarios per model, establish a first robust conclusion: shrinkage estimators systematically dominate Rolling in terms of mean tracking error, on both tested universes, with a hierarchy QIS > LW > ANLS > Rolling.

2.2 Effect of Data Frequency

Table 14: Aggregation by data frequency — MSDEEMUN

Frequency	$N_{\text{scenarios}}$	Mean TE	Std TE	Min TE	Max TE
Daily	36 000	1.39%	0.68%	0.16%	3.89%
Weekly	36 000	1.60%	0.81%	0.16%	5.42%

Table 15: Aggregation by data frequency — M1US

Frequency	$N_{\text{scenarios}}$	Mean TE	Std TE	Min TE	Max TE
Daily	36 000	1.56%	0.83%	0.18%	4.37%
Weekly	36 000	1.81%	0.97%	0.18%	4.98%

Table 16: Mean TE by model and data frequency — MSDEEMUN

Model	Daily	Weekly	Δ (weekly − daily)
ANLS	1.43%	1.46%	+0.03%
LW	1.33%	1.43%	+0.10%
QIS	1.30%	1.40%	+0.10%
Rolling	1.52%	2.11%	+0.60%

Table 17: Mean TE by model and data frequency — M1US

Model	Daily	Weekly	Δ (weekly $-$ daily)
ANLS	1.54%	1.66%	+0.12%
LW	1.51%	1.63%	+0.12%
QIS	1.46%	1.60%	+0.14%
Rolling	1.72%	2.35%	+0.63%

The effect of observation frequency is visible but asymmetric across models. At the aggregate level, the mean tracking error increases from 1.39% on daily data to 1.60% on weekly on MSDEEMUN (+21 basis points), and from 1.56% to 1.81% on M1US (+25 basis points). These gaps, though moderate in absolute terms, reflect the mechanical degradation of the effective p/n ratio induced by the reduction in the number of observations at identical calendar window.

The model-level analysis reveals an important structural asymmetry. Shrinkage models absorb the loss of observations well: on MSDEEMUN, ANLS loses only 3 basis points (+0.03%), while LW and QIS each lose 10 basis points. On M1US, the gaps remain contained between 11 and 15 basis points. This result is explained by their spectral regularization mechanism, which partially corrects the bias introduced by the reduction of the p/n ratio and allows maintaining quality estimation even with fewer observations.

Rolling in contrast suffers a much more severe degradation: +0.60% on MSDEEMUN and +0.63% on M1US, a gap six times larger than that of the shrinkage estimators. Without a regularization mechanism, the empirical estimator is directly exposed to the reduction of the effective p/n ratio: fewer observations means a noisier sample matrix, and this noise translates fully into the tracking error. This asymmetric gap between Rolling and the shrinkage estimators constitutes a direct illustration of the advantage of regularization in underdetermined regimes.

These results suggest that it is preferable to maximize the number of observations by remaining on daily frequency, rather than trying to reduce noise from market frictions by switching to weekly. On large-cap liquid universes such as MSDEEMUN and M1US, microstructure frictions are structurally limited, so that the expected benefit of weekly smoothing is small. In contrast, the reduction of the p/n ratio it implies significantly degrades estimation quality, particularly for non-regularized estimators. Increasing n via daily frequency is therefore the most effective lever for improving the conditioning of the estimated covariance matrix.

2.3 Effect of Rebalancing Frequency

The following tables present the aggregated results by rebalancing frequency: monthly (M), quarterly (Q) and weekly (W), over all scenarios (daily and weekly data combined).

Table 18: Aggregation by rebalancing frequency — MSDEEMUN

Frequency	$N_{\text{scen.}}$	Mean TE	Std TE	Min TE	Max TE	Ann. turnover	TE/Turnover
Monthly	24 000	1.50%	0.75%	0.20%	5.42%	2.25	0.0066
Quarterly	24 000	1.50%	0.76%	0.16%	5.05%	1.53	0.0079
Weekly	24 000	1.49%	0.75%	0.17%	5.33%	3.26	0.0043

Table 19: Aggregation by rebalancing frequency — M1US

Frequency	$N_{\text{scen.}}$	Mean TE	Std TE	Min TE	Max TE	Ann. turnover	TE/Turnover
Monthly	24 000	1.71%	0.92%	0.22%	4.98%	2.65	0.0063
Quarterly	24 000	1.68%	0.91%	0.21%	4.83%	1.51	0.0076
Weekly	24 000	1.66%	0.91%	0.18%	4.83%	3.93	0.0043

Table 20: Mean annualized TE by model and rebalancing frequency — MSDEEMUN

Model	Monthly	Quarterly	Weekly
ANLS	1.45%	1.45%	1.43%
LW	1.39%	1.39%	1.37%
QIS	1.36%	1.35%	1.34%
Rolling	1.82%	1.82%	1.80%

Table 21: Mean annualized TE by model and rebalancing frequency — M1US

Model	Monthly	Quarterly	Weekly
ANLS	1.62%	1.60%	1.58%
LW	1.59%	1.57%	1.55%
QIS	1.55%	1.53%	1.51%
Rolling	2.07%	2.01%	2.01%

The results confirm that rebalancing frequency is a secondary parameter: the three frequencies produce near-identical tracking error levels. On MSDEEMUN, the aggregate means are practically indistinguishable (1.49% to 1.50%), and on M1US the maximum gap between frequencies is only 5 basis points (1.66% to 1.71%). The model-level analysis confirms this conclusion: QIS moves from 1.36% monthly to 1.34% weekly on MSDEEMUN, a gain of only 2 basis points. The gaps are comparable for LW and ANLS.

Turnover, however, remains highly differentiated by frequency. On MSDEEMUN, it moves from 1.53 quarterly to 3.26 weekly, more than double. On M1US, the gap is even more pronounced: from 1.51 to 3.93. The TE per unit of turnover metric confirms that quarterly frequency is the most efficient on both universes (0.0079 on MSDEEMUN, 0.0076 on M1US), achieving near-identical tracking error to weekly frequency while mobilizing twice less turnover. Weekly frequency, with a ratio of 0.0043 on both universes, requires the most turnover per basis point of TE gained and is therefore the least efficient transactionally. Assuming zero transaction costs, the marginal gain of weekly frequency (1 to 5 basis points depending on the universe) does not justify the doubling of turnover it implies. With the introduction of realistic costs, this gain would likely be eliminated.

The model hierarchy remains perfectly stable regardless of rebalancing frequency: QIS and LW systematically dominate, followed by ANLS, then Rolling. This result confirms that rebalancing frequency does not affect the relative ranking of the estimators.

2.4 Effect of the Data Window

The following tables analyze the impact of estimation window length on tracking error, first in an aggregated manner across all models, then by crossing window, exclusion rate, and model.

Table 22: Aggregation by data window — MSDEEMUN

Window (days)	$N_{\text{scen.}}$	Mean TE	Std TE	Min TE	Max TE
126	14 400	1.56%	0.76%	0.16%	4.04%
252	14 400	1.55%	0.78%	0.16%	5.19%
378	14 400	1.51%	0.79%	0.17%	5.42%
504	14 400	1.45%	0.73%	0.16%	4.25%
756	14 400	1.42%	0.70%	0.17%	3.59%

Table 23: Aggregation by data window — M1US

Window (days)	$N_{\text{scen.}}$	Mean TE	Std TE	Min TE	Max TE
126	14 400	1.84%	1.00%	0.18%	4.98%
252	14 400	1.72%	0.92%	0.18%	4.89%
378	14 400	1.66%	0.89%	0.18%	4.83%
504	14 400	1.62%	0.87%	0.18%	4.61%
756	14 400	1.57%	0.85%	0.18%	4.20%

Table 24: Mean TE by model and window — MSDEEMUN

Model	126d	252d	378d	504d	756d	Δ best/worst
ANLS	1.47%	1.49%	1.42%	1.41%	1.44%	0.08%
LW	1.43%	1.40%	1.38%	1.36%	1.34%	0.09%
QIS	1.39%	1.36%	1.35%	1.34%	1.32%	**0.07%**
Rolling	1.94%	1.96%	1.87%	1.72%	1.59%	0.35%

Table 25: Mean TE by model and window — M1US

Model	126d	252d	378d	504d	756d	Δ best/worst
ANLS	1.69%	1.61%	1.59%	1.58%	1.53%	0.16%
LW	1.63%	1.58%	1.56%	1.54%	1.53%	0.10%
QIS	1.64%	1.55%	1.51%	1.49%	1.47%	**0.17%**
Rolling	2.41%	2.13%	1.97%	1.88%	1.76%	0.65%

Table 26: Mean TE — Window × Exclusion rate, all models — MSDEEMUN

Window	$f = 5\%$	$f = 10\%$	$f = 20\%$	$f = 30\%$	$f = 40\%$	$f = 50\%$	Mean
126	0.68%	0.93%	1.43%	1.74%	2.06%	2.50%	1.56%
252	0.73%	0.93%	1.38%	1.70%	2.04%	2.52%	1.55%
378	0.66%	0.87%	1.34%	1.68%	2.02%	2.45%	1.51%
504	0.63%	0.84%	1.32%	1.63%	1.94%	2.37%	1.45%
756	0.62%	0.83%	1.28%	1.59%	1.89%	2.32%	1.42%
Mean	0.66%	0.88%	1.35%	1.67%	1.99%	2.43%	1.50%

Table 27: Mean TE — Window × Exclusion rate, all models — M1US

Window	$f = 5\%$	$f = 10\%$	$f = 20\%$	$f = 30\%$	$f = 40\%$	$f = 50\%$	Mean
126	0.79%	1.16%	1.70%	2.07%	2.41%	2.94%	1.84%
252	0.72%	1.08%	1.59%	1.93%	2.25%	2.75%	1.72%
378	0.67%	1.02%	1.53%	1.88%	2.18%	2.68%	1.66%
504	0.64%	0.99%	1.50%	1.84%	2.14%	2.62%	1.62%
756	0.60%	0.96%	1.46%	1.80%	2.08%	2.54%	1.57%
Mean	0.68%	1.04%	1.55%	1.90%	2.21%	2.71%	1.68%

Table 28: Mean TE by model and exclusion fraction — MSDEEMUN

Model	$f = 5\%$	$f = 10\%$	$f = 20\%$	$f = 30\%$	$f = 40\%$	$f = 50\%$
ANLS	0.61%	0.83%	1.30%	1.62%	1.94%	2.38%
LW	0.55%	0.76%	1.23%	1.55%	1.88%	2.33%
QIS	0.53%	0.74%	1.20%	1.52%	1.84%	2.29%
Rolling	0.97%	1.20%	1.68%	1.99%	2.31%	2.74%

Table 29: Mean TE by model and exclusion fraction — M1US

Model	$f = 5\%$	$f = 10\%$	$f = 20\%$	$f = 30\%$	$f = 40\%$	$f = 50\%$
ANLS	0.56%	0.94%	1.47%	1.82%	2.15%	2.66%
LW	0.54%	0.92%	1.44%	1.80%	2.11%	2.60%
QIS	0.52%	0.88%	1.40%	1.75%	2.06%	2.57%
Rolling	1.10%	1.42%	1.90%	2.24%	2.53%	3.00%

Window lengthening improves mean tracking error monotonically on both universes. On MSDEEMUN, it decreases from 1.56% at 126 days to 1.42% at 756 days, a gain of 14 basis points. On M1US, the gain is more pronounced: from 1.84% at 126 days to 1.57% at 756 days, i.e., 27 basis points. The trend is regular and shows no plateau on M1US, indicating that window lengthening continues to bring a benefit even at 756 days on this large-dimensional universe.

The model-level analysis confirms highly differentiated behaviors in response to window lengthening. On MSDEEMUN, shrinkage estimators present low Δ best/worst differentials: 0.07% for QIS, 0.09% for LW and 0.08% for ANLS, reflecting low sensitivity to window length. Their regularization mechanism effectively compensates for the bias related to the p/n ratio even on short windows. Rolling presents conversely a differential of 0.35%, confirming its strong exposure to the p/n ratio: it moves from 1.94% at 126 days to 1.59% at 756 days, a notable improvement that nevertheless does not allow it to reach shrinkage levels even at long windows.

On M1US, the effect is even more pronounced for Rolling: its differential reaches 0.65%, moving from 2.41% at 126 days to 1.76% at 756 days, the largest gain observed among all models and universes. QIS and LW maintain contained differentials (0.17% and 0.10%), confirming their structural robustness. A notable result on M1US concerns ANLS, whose optimal window is 756 days with a level of 1.53%, identical to LW on this horizon, suggesting a convergence of the nonlinear and linear shrinkage approaches when the number of observations is sufficient.

The crossed window $\times$ exclusion rate table confirms that the beneficial effect of lengthening is robust at all exclusion levels. The 756-day window systematically produces the lowest tracking error regardless of the excluded fraction. Furthermore, the exclusion rate remains the most determinant factor in absolute level: on MSDEEMUN, the mean TE moves from 0.66% at $f = 5\%$ to 2.43% at $f = 50\%$, a factor of nearly 4. On M1US, this factor is even more pronounced: from 0.68% to 2.71%.

The model $\times$ exclusion table reveals that the model hierarchy is rigorously preserved at all exclusion levels. On MSDEEMUN, QIS dominates at all levels with TE values between 0.53% ($f = 5\%$) and 2.29% ($f = 50\%$), versus 0.97% and 2.74% for Rolling. The gap between QIS and Rolling amplifies with the exclusion rate: from 44 basis points at $f = 5\%$ to 45 basis points at $f = 50\%$, confirming that the shrinkage advantage is persistent regardless of the constraint level. On M1US, the same hierarchy is observed with even more pronounced gaps between Rolling and the shrinkage estimators (58 basis points at $f = 5\%$, 43 basis points at $f = 50\%$).

In summary, these results empirically illustrate the sensitivity of the empirical estimator to the p/n ratio: Rolling benefits strongly from window lengthening, while the shrinkage estimators, by regularizing the estimation, are structurally less dependent on this parameter. QIS maintains a stable advantage across all configurations, regardless of the exclusion rate and window retained.

2.5 Robustness and Stress Behavior

2.5.1 Bear Market Behavior and Temporal Stability

Table 30: Risk profile by model — MSDEEMUN

Model	Mean TE	Bear TE	Bear/global ratio	TE stability	Mean Max DD
ANLS	1.45%	1.40%	0.964	0.290%	-4.88%
LW	1.38%	1.34%	0.971	0.280%	-4.72%
QIS	1.35%	1.31%	0.969	**0.270%**	-4.78%
Rolling	1.81%	1.75%	0.967	0.400%	**-5.65%**

Table 31: Risk profile by model — M1US

Model	Mean TE	Bear TE	Bear/global ratio	TE stability	Mean Max DD
ANLS	1.60%	1.58%	0.985	0.310%	-5.28%
LW	1.57%	1.54%	0.982	0.300%	-4.97%
QIS	1.53%	1.51%	0.986	**0.290%**	**-5.04%**
Rolling	2.03%	2.00%	0.985	0.410%	-6.11%

Bear TE denotes the annualized tracking error computed solely on days when the benchmark return is negative. The bear/global ratio measures whether a model deteriorates more during stress periods than over the full period.

Two findings emerge. First, all models see their TE slightly decrease in a bear market relative to their global mean (ratio below 1), which is explained by the very nature of the TE optimizer: during downturns, correlations between assets increase and converge, which makes the covariance matrix easier to estimate and optimization more stable. Second, the ratios are remarkably tight across models on both universes (0.964 to 0.971 on MSDEEMUN, 0.982 to 0.986 on M1US), suggesting that all estimators react in a near-identical manner to stress periods in relative terms. Differentiation occurs primarily on the absolute TE level, where the hierarchy QIS $<$ LW $<$ ANLS $<$ Rolling is fully preserved.

TE stability, defined as the standard deviation of annual tracking errors over the period, measures the temporal regularity of replication. QIS displays the highest stability on both universes (0.270% on MSDEEMUN, 0.290% on M1US), followed by LW (0.280% and 0.300%). Rolling presents the lowest stability on both universes (0.400% on MSDEEMUN, 0.410% on

M1US), reflecting greater year-to-year variability linked to the absence of regularization and direct sensitivity to the p/n ratio.

The mean active drawdown confirms the hierarchy: Rolling presents the most severe drawdowns on both universes (-5.65% on MSDEEMUN, -6.11% on M1US), reflecting structurally stronger exposure to prolonged relative loss episodes. QIS and LW display the least severe drawdowns, with comparable levels between them (-4.72% to -4.88% on MSDEEMUN, -4.97% to -5.04% on M1US).

2.5.2 Inter-Seed Stability

Inter-seed stability measures the extent to which results depend on the random draw of excluded universes. For each model, the standard deviation of annualized TE across the 100 seeds used per exclusion level is computed, as well as the coefficient of variation $CV = \sigma_{TE}/\mu_{TE}$.

Table 32: Inter-seed stability — MSDEEMUN and M1US

	MSDEEMUN		M1US	
Model	σ_{TE} (seeds)	CV	σ_{TE} (seeds)	CV
ANLS	0.71%	0.491	0.89%	0.558
LW	0.71%	0.514	0.88%	0.558
QIS	0.70%	0.517	0.87%	0.567
Rolling	0.81%	0.445	0.92%	0.454

The coefficients of variation are comparable across models on each universe, with no estimator standing out by a particularly high or low sensitivity to the draw of excluded universes. Shrinkage estimators present slightly higher CV values than Rolling (0.491–0.517 versus 0.445 on MSDEEMUN), which is explained by their lower TE level: at comparable absolute variability, a higher CV simply reflects a smaller denominator (μ_{TE}). This result indicates that the variability observed between scenarios primarily reflects the variability of the generated investable universes rather than any instability intrinsic to the models, and that the choice of excluded assets is at least as determinant as the choice of estimation model for explaining the tracking error level of a given scenario.

It is also noted that CV values are slightly higher on M1US than on MSDEEMUN (0.454–0.567 versus 0.445–0.517), reflecting greater heterogeneity in the configurations generated by random exclusions on this larger index.

3 General Summary and Lessons from the Empirical Validation

The evaluation conducted on 72 000 scenarios per index, systematically varying the operational parameters (data frequency, rebalancing frequency, estimation window, exclusion rate), allows us to draw robust lessons on the behavior of covariance estimation models in a broader framework less dependent on a particular trajectory. This phase excludes the EWMA model, whose structural unsuitability for the high-dimensional replication framework was established in the previous parts.

1. A stable model hierarchy. The first major lesson of this validation is the stability of the model hierarchy across all tested configurations. Regardless of the parameter combination retained — data frequency, estimation window, exclusion rate, or rebalancing frequency — the following ranking is observed consistently on both universes:

$$\text{QIS} \ > \ \text{LW} \ > \ \text{ANLS} \ > \ \text{Rolling}.$$

QIS systematically dominates on both universes and in all tested configurations, followed by LW with a gap of 3 to 7 basis points depending on the configuration. ANLS sits between the first two shrinkage estimators and Rolling, retaining a structural advantage over the empirical estimator but without reaching the level of LW and QIS. This stability is all the more remarkable as the absolute tracking error levels vary considerably across configurations: from less than 0.5% in lightly constrained universes to more than 3% in heavily excluded universes. The relative ranking resists these amplitude variations, which constitutes a robust validation of the superiority of shrinkage models over the empirical approach.

2. The exclusion rate as the primary explanatory factor. Among all tested parameters, the exclusion rate is by far the most determinant of the absolute tracking error level. On MSDEEMUN, the mean TE moves from 0.66% at $f = 5\%$ to 2.43% at $f = 50\%$, a factor of nearly 4, regardless of the model or window retained. On M1US, this factor is even more pronounced (from 0.68% to 2.71%). This result confirms that the constraint on the investable universe is the primary driver of tracking error in practice. The choice of estimation model acts in second rank, modulating the way in which the optimizer exploits the available degrees of freedom in a given constrained universe.

3. Data frequency: daily preferable at large dimension. On both tested large-cap universes, daily data systematically produce better performance than weekly data. For shrinkage models, the degradation from switching to weekly remains moderate, from 3 to 15 basis points depending on the model and universe, as their regularization mechanism partially compensates for the reduction of the effective p/n ratio induced by the scarcity of observations. Rolling in contrast suffers a much more pronounced degradation (+0.60% on MSDEEMUN, +0.63% on M1US), confirming its direct exposure to the p/n ratio. On liquid universes listed on the same exchange, microstructure frictions are structurally limited, so that the expected benefit of weekly smoothing is negligible relative to the cost of reducing the number of observations. Maximizing n via daily frequency is therefore the most effective lever for improving the conditioning of the estimated matrix, in particular for non-regularized estimators.

4. The estimation window as a differentiated lever across models. Window lengthening improves tracking error monotonically on both universes, but its effects are very unequal across models. Rolling is the most sensitive to this parameter: on M1US, it gains 27 basis points between 126 and 756 days, confirming its direct dependence on the p/n ratio. QIS and LW present much smaller differentials, demonstrating that their regularization mechanism effectively compensates for estimation bias even under short windows. This result constitutes a direct empirical validation of the theoretical objective of shrinkage estimators: reducing estimation uncertainty in unfavorable regimes, precisely where empirical approaches are most exposed.

5. Rebalancing frequency as a secondary parameter. In the absence of transaction costs, rebalancing frequency has only a marginal impact on tracking error: the maximum gap between frequencies is 1 basis point on MSDEEMUN and 5 basis points on M1US. In contrast, the generated turnover varies by a factor of two between quarterly and weekly frequency. Quarterly frequency emerges as the best TE/turnover trade-off on both universes. With realistic transaction costs, weekly frequency would quickly be penalized, making it a parameter to evaluate jointly with operational costs rather than in isolation.

6. Stress robustness and temporal stability. The analysis of bear market behavior confirms that all models see their tracking error slightly decrease during stress periods, a phenomenon linked to correlation convergence that simplifies the estimation problem. The bear/global ratios are remarkably homogeneous across models (0.964–0.971 on MSDEEMUN, 0.982–0.986 on M1US), indicating that differentiation occurs primarily on the absolute TE level and not on relative stress sensitivity. In terms of temporal stability, QIS presents the lowest TE stability on both universes, confirming that analytical shrinkage produces more regular estimates from year to year. Rolling is the least stable on both universes, with the most severe mean active drawdowns (-5.65% on MSDEEMUN, -6.11% on M1US).

7. A robust and generalizable conclusion. The empirical validation confirms and generalizes the lessons from the economic backtest. The hierarchy QIS > LW > ANLS > Rolling is observed consistently across 72 000 scenarios per index, covering a wide variety of operational contexts. This robustness is a sign that the observed differences are not the product of a particular configuration but genuinely reflect structural properties of the estimators. Shrinkage models, particularly QIS and LW, draw their advantage from their ability to correct estimation bias in unfavorable regimes (high p/n, short window, constrained universe), precisely the conditions most frequently encountered in practical index replication applications.

General Conclusion

Main Findings

This study compared five covariance matrix estimators — Rolling sample, EWMA, Ledoit-Wolf linear shrinkage (LW), analytical nonlinear shrinkage (ANLS), and Quadratic-Inverse Shrinkage (QIS) — along two complementary axes: a statistical evaluation in a controlled environment where the true covariance is known, and an economic evaluation via index replication backtests on real data, extended by an empirical validation on 90 000 scenarios.

QIS and LW dominate, with a stable and robust hierarchy. The most salient result of this study is the consistency and stability of the model hierarchy across all evaluation frameworks, tested configurations, and considered universes:

$$\text{QIS} > \text{LW} > \text{ANLS} > \text{Rolling} > \text{EWMA}.$$

QIS establishes itself as the most robust estimator in the static i.i.d. framework (Static Oracle), where it dominates on the Stein loss in all p/T regimes and under both innovation types, with an advantage that amplifies as dimension increases relative to the sample. In the dynamic framework (Factor Shock), LW takes a slight advantage over QIS under Gaussian innovation, benefiting from the strong persistence of the covariance. In the economic backtest and empirical validation, QIS systematically dominates on short windows and constrained universes, while LW approaches on long windows.

The p/n ratio as a unified reading key. All results organize around a single principle: the p/n ratio is the determining factor of estimation quality. When p/n is high (short window, large dimension), non-regularized estimators (Rolling, EWMA) degrade strongly, while shrinkage estimators maintain stable performance. Window lengthening improves all models but benefits Rolling more, whose relative disadvantage versus shrinkage models decreases mechanically as n increases. This result empirically validates the theoretical framework that motivates the construction of QIS and LW.

EWMA, unsuited to large universes. EWMA is structurally the least performant across all configurations. In the i.i.d. framework, its exponential weighting mechanism is poorly adapted to a fixed covariance. In the large-dimensional dynamic framework, its Stein loss explodes. In the backtest, it systematically displays the highest tracking error, least sensitive to window lengthening. This result suggests that EWMA, designed for restricted universes and highly reactive covariances, is not suited to the index replication framework on large universes.

Data frequency: daily preferable at large dimension. On both tested universes, daily data systematically produce better performance than weekly data. This result is explained by the fact that switching to weekly mechanically reduces the effective p/n ratio, degrading the conditioning of the estimated covariance matrix, without the reduction in microstructure noise compensating for this loss. On large-cap liquid universes listed on the same exchange, microstructure frictions are structurally limited: closing prices are synchronous and non-synchronicity inference biases are small. Maximizing the number of observations via daily frequency is therefore the most effective lever for improving estimation quality, in particular for non-regularized estimators whose performance is directly exposed to the p/n ratio.

This conclusion is however conditional on the nature of the studied universe. On a multi-asset class or multi-geography universe, mixing for example equities and bonds, or securities listed on distinct time zones, daily data could introduce larger non-synchronicity biases, making the argument in favor of weekly data potentially more relevant.

Limitations and Extensions

Several limitations of the study deserve to be highlighted. Each naturally defines a direction for further investigation.

Fixed parameters of the Factor Shock DGP. The statistical evaluation on the Factor Shock DGP relies on a unique calibration of dynamic parameters ($\rho_B = 0.95$, $\sigma_B = 0.05$, $\rho_d = 0.97$, $\sigma_d = 0.10$), corresponding to a strongly persistent regime favorable to long-window estimators and stable shrinkage. A faster non-stationarity regime ($\rho_B \approx 0.5$) could modify the hierarchy. An extended parametric grid — $\rho_B \in \{0.5; 0.8; 0.95; 0.99\}$, $\sigma_B \in \{0.02; 0.05; 0.15\}$, Student degree of freedom $\nu = 3$ instead of $\nu = 5$ — would allow tracing the boundary in parameter space where LW takes the advantage over QIS, and possibly identify the regimes in which EWMA becomes competitive again through its reactivity. The Factor Shock conclusions must therefore be interpreted within this specific parametric context.

Two large-cap equity universes. The economic evaluation covers exclusively two large-cap equity indices (MSDEEMUN and M1US), composed of liquid assets listed on the same reference exchange. This homogeneity structurally reduces microstructure frictions and favors daily data. The absence of a small-size universe, typically 50 assets or fewer, does not allow evaluation of the dimension/frequency trade-off in a regime where the weekly p/n ratio would remain manageable and where the reduction of microstructure noise could compensate for the loss of observations. Adding such a universe would allow determining whether the daily superiority established here, conditional on the homogeneous and liquid nature of the studied universe, is maintained in configurations less favorable to daily data.

Single asset class. The study is limited to listed equities. Less liquid asset classes — bonds, mutual funds, funds of funds, alternative assets — present different covariance structures, naturally weekly or monthly valuation frequencies, shorter histories, and more unstable correlations. In these universes, daily data could introduce non-synchronicity biases and microstructure effects that would make the argument in favor of weekly data more relevant. The transferability of conclusions to these contexts is not guaranteed and constitutes a natural extension.

Absence of a factor model as reference. The five compared estimators are all direct estimators of the covariance matrix. A structural factor model of the Fama-French, APT, or PCA type would constitute a relevant alternative reference, in particular in large-dimensional universes where the underlying factor structure is strong. Adding such an estimator would allow quantifying the marginal gain provided by nonparametric shrinkage relative to an explicit modeling of asset dependence, and evaluating whether QIS retains its advantage over a well-calibrated factor model on large indices.

Appendices

A Parameters of the Statistical Evaluation

Table 33: Parameters common to both DGPs

Parameter	Value
Scenarios K	10 000 (Static Oracle) / 1 000 (Factor Shock)
Base seed	42
Target daily volatility	1%
Number of factors	10
Idiosyncratic weight	0.30
Innovations	Gaussian $\mathcal{N}(0, 1)$ and Student t_5 ($\nu = 5$)

Table 34: Specific parameters — DGP Static Oracle

Parameter	Value
Ratio p/T	0.2 and 1.0
Length T	$T = p / (p/T)$ + burn-in
Burn-in	60 observations
True covariance Σ	One matrix per seed (factor structure + lognormal idio)

Table 35: Specific parameters — DGP Factor Shock

Parameter	Value
N_{sim}	100 and 500 assets
T_{sim}	≈ 805 days (length of R_{ref})
Burn-in	60 observations (excluded from loss computation)
Matrices evaluated / scenario	≈ 553 (252-day window) / ≈ 301 (504-day window)
ρ_B / σ_B	0.95 / 0.05 (factor loadings)
ρ_d / σ_d	0.97 / 0.10 (log-idiosyncratic variances)

Table 36: Models and windows — statistical evaluation

Model	Windows tested
Rolling	252d, 504d
EWMA ($\lambda = 0.94$)	252d, 504d
LW 2004	252d, 504d
ANLS 2020	252d, 504d
QIS 2022	252d, 504d

B Parameters of the Economic Validation

Table 37: Scenario grid — Economic validation

Dimension	Values	Total
Exclusion fractions f	5%, 10%, 20%, 30%, 40%, 50%	6
Seeds per fraction	100	600 universes / index
Rebalancing frequencies	M, Q, W	3
Data windows	126d, 252d, 378d, 504d, 756d	5
Observation frequencies	Daily, weekly	2
Scenarios / model / index	$600 \times 4 \times 2 \times 3$	14 400
Daily scenarios / model	$600 \times 4 \times 3$	7 200
Total all models (1 index)	$18\,000 \times 4$	72 000

Table 38: Models and windows — Economic validation

Model	Windows tested
Rolling	126d, 252d, 378d, 504d, 756d
EWMA ($\lambda = 0.94$)	126d, 252d, 378d, 504d, 756d
LW 2004	126d, 252d, 378d, 504d, 756d
ANLS 2020	126d, 252d, 378d, 504d, 756d
QIS 2022	126d, 252d, 378d, 504d, 756d

Table 39: Indices — Economic validation

Index	Assets	Period
MSDEEMUN (euro zone)	≈ 240	2020–2025
M1US (United States)	≈ 600	2020–2025

C Detailed Annual Results — Economic Evaluation

C.1 MSDEEMUN — Full Universe Without Exclusions

252-Day Window

Table 40: Annual backtest results — MSDEEMUN without exclusions (252-day window, period 2020–2025)

	ANLS 252d		LW 252d		Rolling 252d		EWMA 252d		QIS 252d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	0.08%	0.06%	0.08%	0.06%	0.07%	0.06%	0.41%	0.26%	0.08%	0.06%
2021	0.04%	0.07%	0.02%	0.06%	0.04%	0.07%	0.14%	0.17%	0.03%	0.07%
2022	0.01%	0.15%	0.06%	0.13%	0.00%	0.15%	-0.22%	0.33%	0.07%	0.14%
2023	-0.00%	0.07%	0.00%	0.07%	0.00%	0.07%	-0.19%	0.15%	-0.00%	0.07%
2024	0.01%	0.14%	0.02%	0.11%	0.01%	0.13%	-0.07%	0.18%	0.04%	0.11%
2025	0.02%	0.10%	-0.04%	0.09%	0.02%	0.10%	0.13%	0.16%	-0.05%	0.09%
Cumulative total	0.15%	0.10%	0.16%	0.09%	0.14%	0.10%	0.18%	0.22%	0.17%	0.09%
Annualized	0.03%	0.10%	0.03%	0.09%	0.02%	0.10%	0.03%	0.22%	0.03%	0.09%

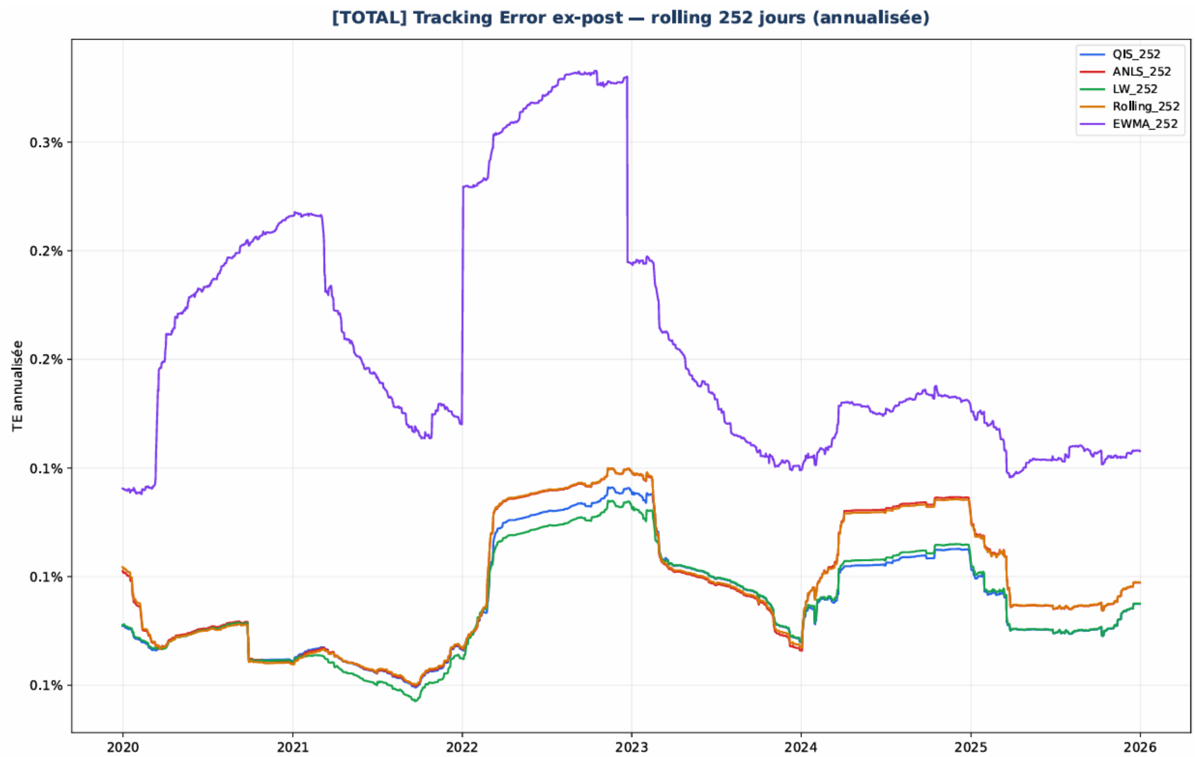

Figure 1: Ex-post tracking error (rolling 252 days) — MSDEEMUN without exclusions (252-day window).

On the 252-day window, tracking error levels are very low and gaps between models remain limited. LW and QIS stand out slightly with values around 0.06%–0.07%, while ANLS and

Rolling sit around 0.07%–0.10%. EWMA, in contrast, presents significantly higher levels (0.17%–0.26%), reflecting increased sensitivity to recent variations.

Stress years, notably 2022, slightly accentuate the gaps: LW and QIS remain the most stable, ANLS and Rolling rise to 0.14%–0.15%, while EWMA reaches 0.33%. The chart confirms this dynamic, with shrinkage curves very close together and a notably more volatile EWMA trajectory.

504-Day Window

Table 41: Annual backtest results — MSDEEMUN without exclusions (504-day window, period 2020–2025)

	ANLS 504d		LW 504d		Rolling 504d		EWMA 504d		QIS 504d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	0.08%	0.06%	0.08%	0.06%	0.08%	0.06%	0.40%	0.26%	0.08%	0.06%
2021	0.03%	0.07%	0.02%	0.06%	0.03%	0.07%	0.14%	0.17%	0.03%	0.07%
2022	0.03%	0.14%	0.05%	0.14%	0.03%	0.14%	-0.22%	0.33%	0.05%	0.14%
2023	-0.00%	0.07%	-0.01%	0.07%	-0.00%	0.07%	-0.19%	0.15%	-0.01%	0.07%
2024	0.05%	0.12%	0.07%	0.12%	0.05%	0.12%	-0.07%	0.18%	0.08%	0.12%
2025	-0.05%	0.09%	-0.05%	0.09%	-0.05%	0.09%	0.13%	0.16%	-0.05%	0.09%
Cumulative total	0.14%	0.10%	0.15%	0.09%	0.14%	0.10%	0.17%	0.22%	0.18%	0.09%
Annualized	0.02%	0.10%	0.03%	0.09%	0.02%	0.10%	0.03%	0.22%	0.03%	0.09%

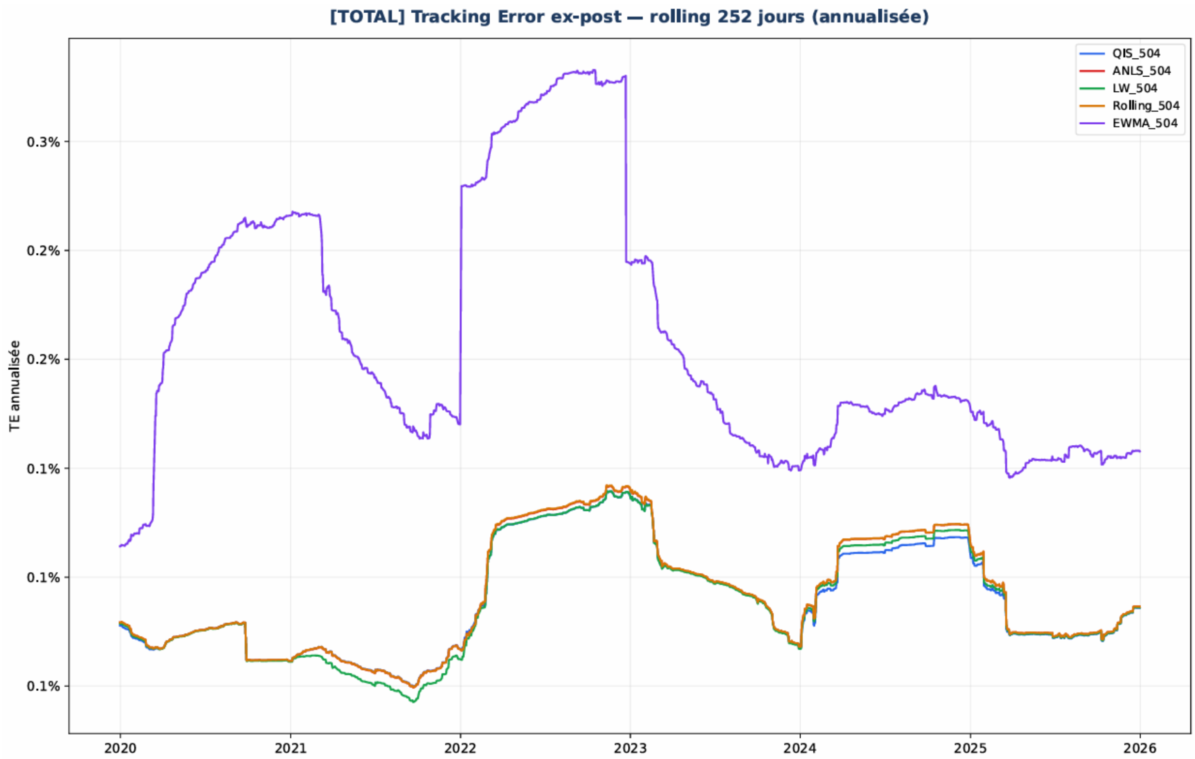

Figure 2: Ex-post tracking error (rolling 252 days) — MSDEEMUN without exclusions (504-day window).

With a 504-day window, tracking error levels become even more homogeneous. LW, QIS and ANLS converge toward 0.06%–0.07% in most years, while Rolling remains slightly above but very close. EWMA retains significantly higher levels (0.17%–0.26%), with marked peaks in 2020 and 2022.

The chart illustrates near-perfect convergence between the shrinkage models and Rolling, while EWMA remains isolated above. Window lengthening therefore reduces estimation noise but does not modify the hierarchy.

756-Day Window

Table 42: Annual backtest results — MSDEEMUN without exclusions (756-day window, period 2020–2025)

	ANLS 756d		LW 756d		Rolling 756d		EWMA 756d		QIS 756d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	0.11%	0.06%	0.11%	0.06%	0.11%	0.06%	0.18%	0.12%	0.11%	0.06%
2021	0.03%	0.07%	0.03%	0.07%	0.03%	0.07%	0.13%	0.17%	0.03%	0.07%
2022	0.05%	0.14%	0.05%	0.14%	0.05%	0.14%	-0.22%	0.33%	0.05%	0.14%
2023	-0.01%	0.07%	-0.01%	0.07%	-0.01%	0.07%	-0.19%	0.15%	-0.01%	0.07%
2024	0.08%	0.12%	0.10%	0.12%	0.08%	0.12%	-0.07%	0.18%	0.10%	0.12%
2025	-0.06%	0.09%	-0.06%	0.08%	-0.06%	0.09%	0.13%	0.16%	-0.06%	0.09%
Cumulative total	0.20%	0.10%	0.22%	0.10%	0.20%	0.10%	-0.05%	0.20%	0.22%	0.10%
Annualized	0.03%	0.10%	0.04%	0.10%	0.03%	0.10%	-0.01%	0.20%	0.04%	0.10%

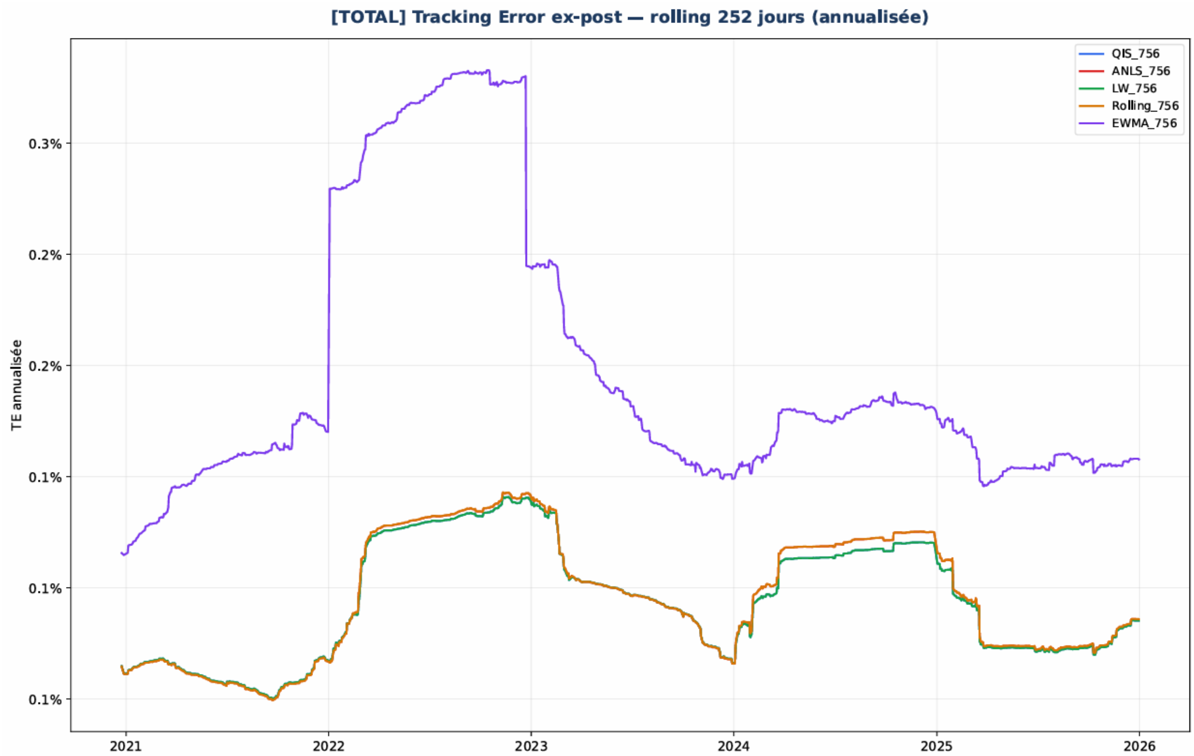

Figure 3: Ex-post tracking error (rolling 252 days) — MSDEEMUN without exclusions (756-day window).

On the 756-day window, tracking error reaches its lowest level and models converge near-perfectly. LW, QIS, ANLS and Rolling all sit around 0.06%–0.10%, with quasi-identical trajectories. Annual variations are minimal, reflecting complete stabilization of covariance estimation.

EWMA remains the only model to diverge, with levels around 0.12%–0.20% and persistent peaks in 2022 and 2025. The chart shows near-perfect superposition of the shrinkage models and Rolling, while EWMA remains clearly above.

C.2 MSDEEMUN — Restricted Universe With Exclusions

252-Day Window

Table 43: Annual results — MSDEEMUN with exclusions (252-day window, period 2020–2025)

	ANLS 252d		LW 252d		Rolling 252d		EWMA 252d		QIS 252d	
Year	AR	TE	AR	TE	AR	TE	AR	TE	AR	TE
2020	1.14%	0.96%	1.14%	0.96%	1.07%	0.95%	0.68%	1.02%	0.35%	0.81%
2021	-0.04%	0.60%	-0.04%	0.60%	0.01%	0.60%	-0.03%	0.63%	0.18%	0.35%
2022	-0.64%	1.12%	-0.64%	1.12%	-0.67%	1.12%	-0.71%	1.43%	-0.60%	0.74%
2023	-1.20%	0.84%	-1.20%	0.84%	-1.19%	0.84%	-0.57%	1.00%	-0.16%	0.54%
2024	0.35%	0.67%	0.35%	0.67%	0.34%	0.67%	-0.19%	0.76%	0.31%	0.47%
2025	-0.02%	0.49%	-0.02%	0.49%	-0.02%	0.49%	-0.07%	0.64%	0.21%	0.40%
Cumulative total	-0.43%	0.81%	-0.43%	0.81%	-0.47%	0.81%	-0.88%	0.95%	0.29%	0.58%
Annualized	-0.07%	0.81%	-0.07%	0.81%	-0.08%	0.81%	-0.15%	0.95%	0.05%	0.58%

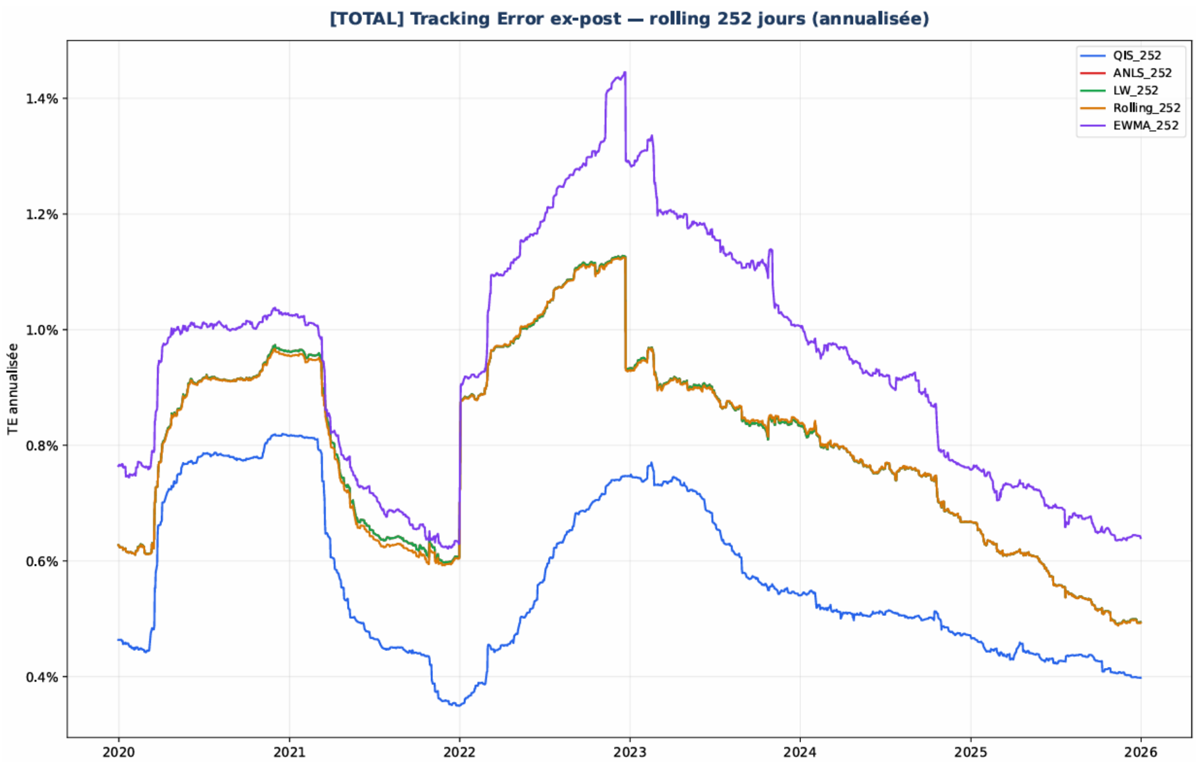

Figure 4: Ex-post tracking error (rolling 252 days) — MSDEEMUN with exclusions (252-day window).

On the 252-day window, tracking error is high and very dispersed, due to a p/n ratio close to 1. QIS stands out clearly with levels around 0.58%, while LW and ANLS sit around 0.81%. Rolling reaches 0.95% and EWMA exceeds 1%, confirming strong sensitivity to estimation errors.

Stress years (2022–2023) amplify these gaps: QIS remains contained (0.74%), while Rolling and EWMA exceed 1.10%. The chart shows more stable shrinkage curves, while Rolling and

EWMA present marked peaks.

504-Day Window

Table 44: Annual results — MSDEEMUN with exclusions (504-day window, period 2020–2025)

	ANLS 504d		LW 504d		Rolling 504d		EWMA 504d		QIS 504d	
Year	AR	TE	AR	TE	AR	TE	AR	TE	AR	TE
2020	0.55%	0.68%	0.67%	0.68%	0.55%	0.68%	0.69%	1.02%	0.68%	0.68%
2021	0.12%	0.39%	0.07%	0.32%	0.12%	0.39%	-0.03%	0.63%	0.04%	0.33%
2022	-0.76%	0.79%	-0.71%	0.73%	-0.76%	0.79%	-0.71%	1.43%	-0.70%	0.74%
2023	-0.46%	0.61%	-0.00%	0.50%	-0.46%	0.61%	-0.57%	1.00%	-0.20%	0.51%
2024	-0.13%	0.52%	0.05%	0.49%	-0.13%	0.52%	-0.19%	0.76%	0.25%	0.47%
2025	0.41%	0.35%	0.33%	0.32%	0.41%	0.35%	-0.07%	0.64%	0.43%	0.33%
Cumulative total	-0.27%	0.58%	0.40%	0.53%	-0.27%	0.58%	-0.88%	0.95%	0.50%	0.54%
Annualized	-0.05%	0.58%	0.07%	0.53%	-0.05%	0.58%	-0.15%	0.95%	0.08%	0.54%

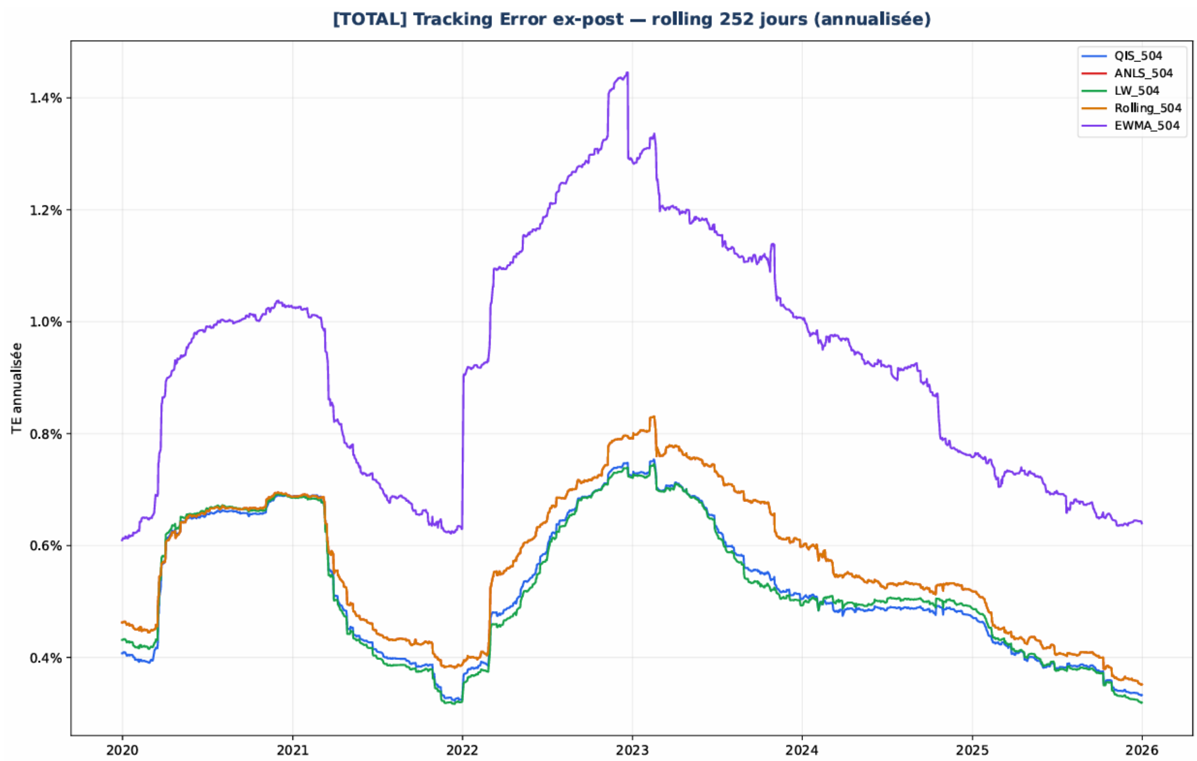

Figure 5: Ex-post tracking error (rolling 504 days) — MSDEEMUN with exclusions (504-day window).

With a 504-day window, tracking error decreases for all models, but gaps remain visible. QIS and LW are the best-performing models (0.53%–0.54%), while ANLS and Rolling sit around 0.58%. EWMA remains clearly above (0.95%).

Years 2020, 2022 and 2024 show that QIS better absorbs stress episodes, LW follows closely, ANLS remains intermediate, and Rolling reacts more strongly. EWMA retains high volatility, with peaks exceeding 1%.

756-Day Window

Table 45: Annual results — MSDEEMUN with exclusions (756-day window, period 2020–2025)

	ANLS 756d		LW 756d		Rolling 756d		EWMA 756d		QIS 756d	
Year	AR	TE	AR	TE	AR	TE	AR	TE	AR	TE
2020	0.46%	0.70%	0.57%	0.69%	0.46%	0.70%	0.05%	0.79%	0.57%	0.69%
2021	0.25%	0.35%	0.18%	0.33%	0.25%	0.35%	-0.03%	0.63%	0.18%	0.33%
2022	-0.69%	0.70%	-0.67%	0.68%	-0.69%	0.70%	-0.71%	1.43%	-0.67%	0.68%
2023	-0.29%	0.55%	-0.16%	0.50%	-0.29%	0.55%	-0.57%	1.00%	-0.16%	0.50%
2024	-0.29%	0.51%	0.03%	0.48%	-0.29%	0.51%	-0.19%	0.76%	0.03%	0.48%
2025	0.19%	0.34%	0.28%	0.32%	0.19%	0.34%	-0.07%	0.64%	0.28%	0.32%
Cumulative total	-0.36%	0.54%	0.22%	0.52%	-0.36%	0.54%	-1.50%	0.92%	0.22%	0.52%
Annualized	-0.06%	0.54%	0.04%	0.52%	-0.06%	0.54%	-0.25%	0.92%	0.04%	0.52%

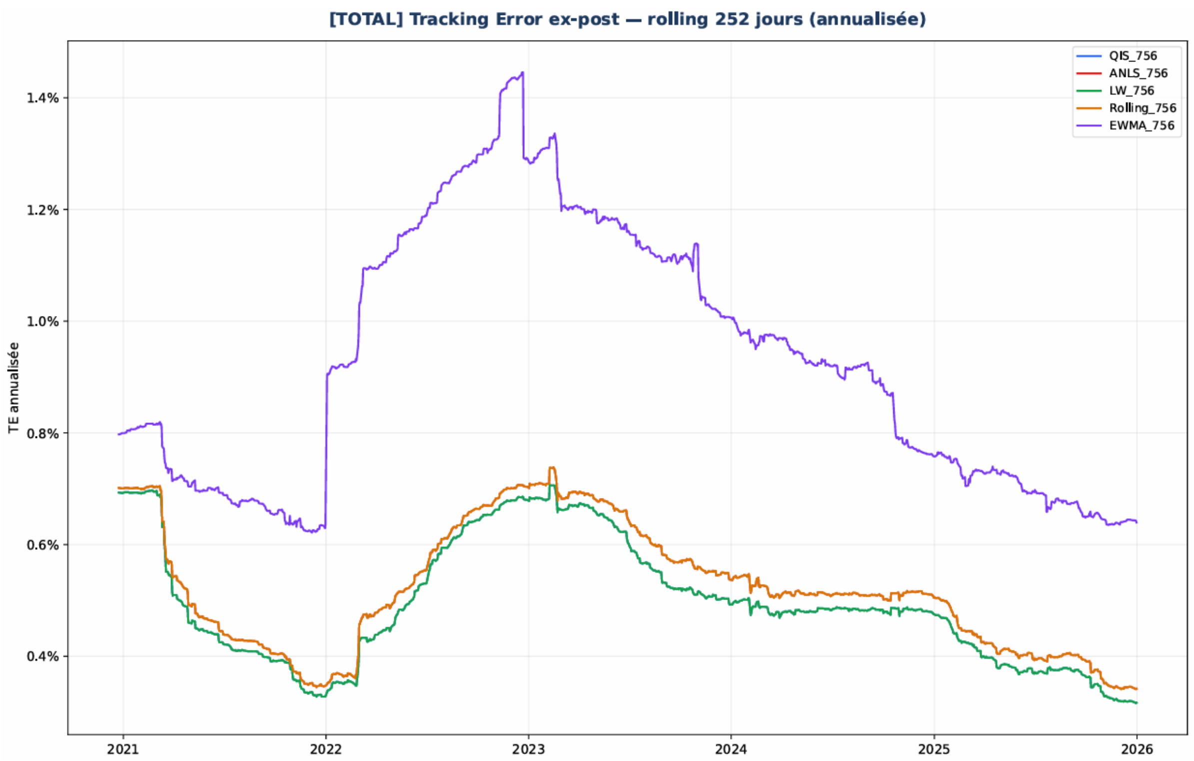

Figure 6: Ex-post tracking error (rolling 756 days) — MSDEEMUN with exclusions (756-day window).

On the 756-day window, tracking error continues to decrease and models converge further. QIS and LW remain the best-performing (0.52%), ANLS and Rolling sit around 0.54%, while EWMA remains clearly above (0.92%).

Stress years confirm this hierarchy: QIS and LW effectively stabilize the covariance, ANLS remains slightly more sensitive, Rolling reacts more to shocks, and EWMA systematically amplifies variations.

C.3 M1US — Full Universe Without Exclusions

252-Day Window

Table 46: Annual results — M1US without exclusions (252-day window)

	ANLS 252d		LW 252d		Rolling 252d		EWMA 252d		QIS 252d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	0.23%	0.14%	0.24%	0.14%	0.55%	0.41%	1.14%	0.51%	0.23%	0.14%
2021	0.04%	0.20%	0.02%	0.20%	0.03%	0.37%	0.12%	0.36%	0.02%	0.20%
2022	0.18%	0.18%	0.18%	0.18%	0.26%	0.33%	0.23%	0.39%	0.18%	0.18%
2023	0.04%	0.07%	0.04%	0.07%	-0.31%	0.31%	-0.43%	0.34%	0.04%	0.07%
2024	0.02%	0.08%	0.02%	0.08%	0.18%	0.26%	0.17%	0.26%	0.01%	0.08%
2025	-0.02%	0.27%	-0.01%	0.27%	-0.17%	0.37%	-0.17%	0.40%	-0.01%	0.27%
Cumulative total	0.49%	0.17%	0.49%	0.17%	0.55%	0.34%	1.07%	0.39%	0.47%	0.17%
Annualized	0.08%	0.17%	0.08%	0.17%	0.09%	0.34%	0.18%	0.39%	0.08%	0.17%

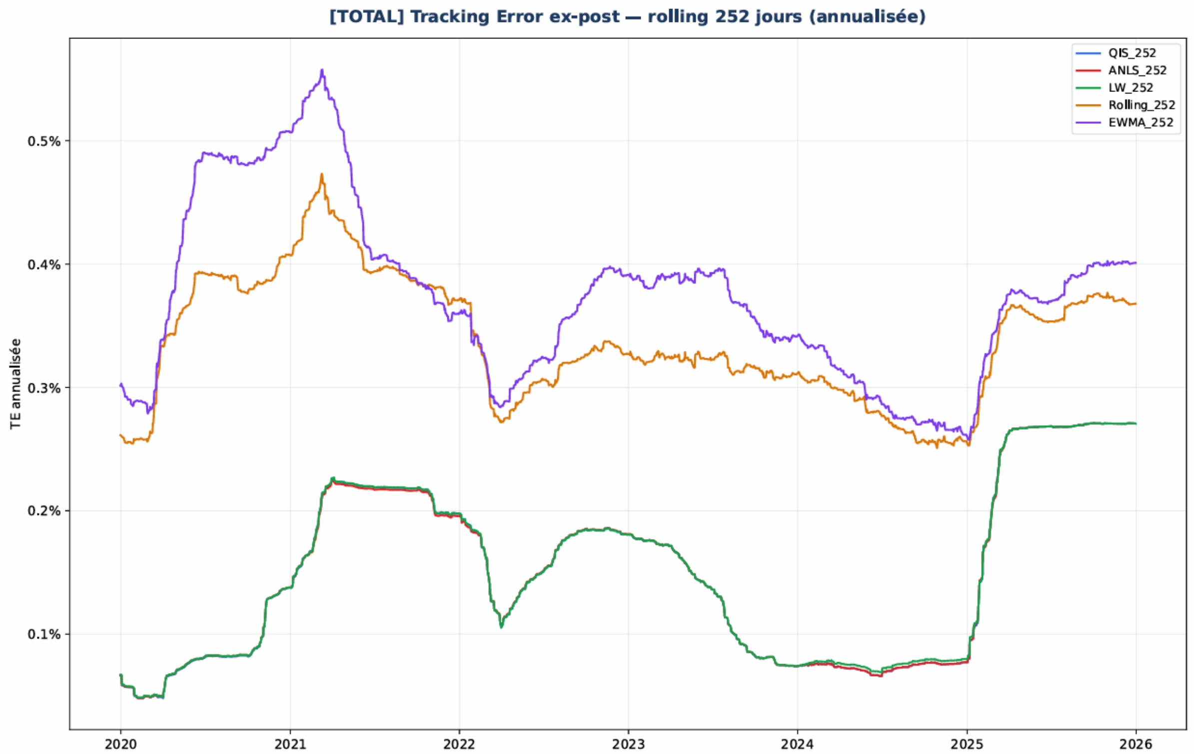

Figure 7: Ex-post tracking error (rolling 252 days) — M1US without exclusions (252-day window).

On the 252-day window, tracking error is higher and more dispersed than in MSDEEMUN, due to a p/n ratio close to 2 in a large-dimensional universe. The shrinkage models stand out clearly: ANLS, LW and QIS sit around 0.14%–0.20%, with near-identical results in 2020, 2021 and 2022. QIS presents slightly superior stability during stress years, while LW shows somewhat higher sensitivity in 2021.

Rolling and EWMA display significantly higher levels, regularly exceeding 0.30% and reaching 0.41%–0.51% in 2020. Years 2022 and 2023 accentuate these gaps: shrinkage models remain contained, while Rolling and EWMA react strongly to market variations. The chart confirms this dynamic, with shrinkage curves very close and marked peaks for Rolling and EWMA.

504-Day Window

Table 47: Annual results — M1US without exclusions (504-day window)

	ANLS 504d		LW 504d		Rolling 504d		EWMA 504d		QIS 504d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	0.24%	0.14%	0.23%	0.14%	0.28%	0.17%	1.16%	0.53%	0.23%	0.14%
2021	0.05%	0.20%	0.03%	0.20%	-0.03%	0.25%	0.12%	0.36%	0.03%	0.19%
2022	0.20%	0.18%	0.19%	0.18%	0.20%	0.20%	0.23%	0.39%	0.18%	0.18%
2023	0.04%	0.07%	0.04%	0.07%	0.00%	0.09%	-0.43%	0.34%	0.04%	0.07%
2024	0.05%	0.08%	0.05%	0.08%	0.03%	0.11%	0.17%	0.26%	0.03%	0.08%
2025	-0.02%	0.27%	-0.02%	0.27%	-0.12%	0.28%	-0.17%	0.40%	-0.02%	0.26%
Cumulative total	0.54%	0.17%	0.53%	0.17%	0.37%	0.19%	1.09%	0.39%	0.50%	0.17%
Annualized	0.09%	0.17%	0.09%	0.17%	0.06%	0.19%	0.18%	0.39%	0.08%	0.17%

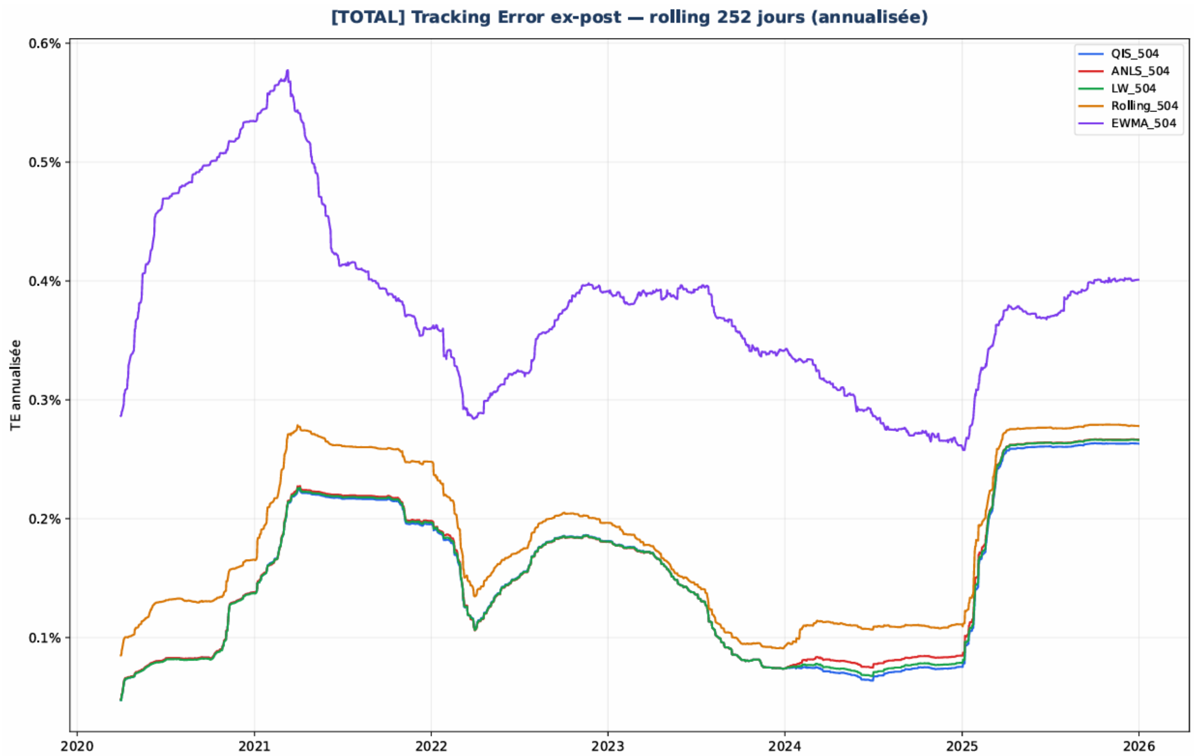

Figure 8: Ex-post tracking error (rolling 252 days) — M1US without exclusions (504-day window).

With a 504-day window, tracking error decreases for all models, reflecting the stabilizing effect of increasing the number of observations. The shrinkage models converge toward a very homogeneous level (0.17%), with near-identical trajectories over the full period. ANLS and LW remain slightly more stable than QIS in 2024, but the gaps remain small.

Rolling also benefits from window lengthening, with tracking error brought down to 0.19%, but remains systematically above the shrinkage models. EWMA retains significantly higher levels (0.39%), with visible peaks in 2020 and 2025. The chart illustrates clear convergence of the shrinkage models, while Rolling and EWMA retain more erratic trajectories.

756-Day Window

Table 48: Annual results — M1US without exclusions (756-day window)

	ANLS 756d		LW 756d		Rolling 756d		EWMA 756d		QIS 756d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	0.03%	0.21%	0.03%	0.21%	0.03%	0.21%	0.20%	0.27%	0.03%	0.21%
2021	0.06%	0.20%	0.03%	0.19%	0.05%	0.20%	0.13%	0.38%	0.03%	0.19%
2022	0.21%	0.18%	0.18%	0.18%	0.22%	0.18%	0.22%	0.39%	0.18%	0.18%
2023	0.04%	0.07%	0.04%	0.07%	0.04%	0.07%	-0.43%	0.34%	0.04%	0.07%
2024	0.02%	0.07%	0.02%	0.07%	0.08%	0.15%	0.17%	0.26%	0.02%	0.07%
2025	-0.07%	0.27%	-0.03%	0.26%	-0.07%	0.28%	-0.17%	0.40%	-0.03%	0.26%
Cumulative total	0.27%	0.18%	0.28%	0.18%	0.28%	0.19%	0.13%	0.35%	0.28%	0.18%
Annualized	0.05%	0.18%	0.05%	0.18%	0.05%	0.19%	0.02%	0.35%	0.05%	0.18%

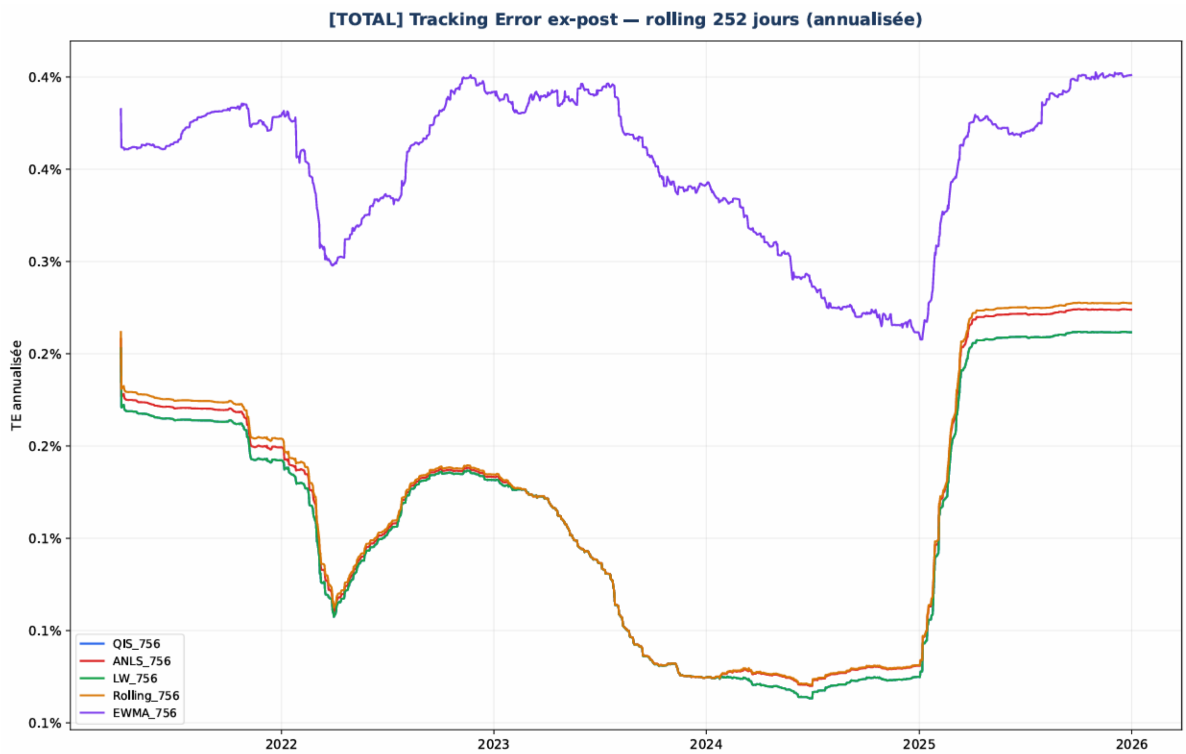

Figure 9: Ex-post tracking error (rolling 252 days) — M1US without exclusions (756-day window).

On the 756-day window, tracking error further decreases and trajectories converge strongly. Shrinkage models (ANLS, LW, QIS) present near-identical levels around 0.18%, with remarkable stability across all years. ANLS and LW retain a slight advantage in 2024, while QIS remains the most regular during stress years.

Rolling approaches the shrinkage models, with tracking error of 0.19%, but remains slightly above. EWMA stays behind, with tracking error of 0.35% and persistent peaks in 2022 and 2025. The chart shows near-superposition of the shrinkage and Rolling curves, while EWMA remains clearly above.

C.4 M1US — Restricted Universe With Exclusions

252-Day Window

Table 49: Annual results — M1US with exclusions (252-day window, period 2020–2025)

	ANLS 252d		LW 252d		Rolling 252d		EWMA 252d		QIS 252d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	1.21%	0.68%	0.93%	0.72%	0.41%	1.03%	1.26%	1.02%	0.97%	0.66%
2021	-0.19%	0.41%	-0.16%	0.42%	0.35%	0.68%	0.54%	0.64%	-0.17%	0.40%
2022	-0.30%	0.62%	-0.77%	0.66%	-0.77%	0.96%	-0.33%	0.82%	-0.63%	0.63%
2023	0.35%	0.57%	0.41%	0.62%	0.19%	1.02%	-0.20%	0.83%	0.62%	0.54%
2024	0.96%	0.41%	1.35%	0.45%	1.52%	0.69%	1.03%	0.57%	1.07%	0.42%
2025	-0.15%	0.51%	0.09%	0.51%	0.24%	0.73%	-0.18%	0.70%	0.11%	0.50%
Cumulative total	1.88%	0.53%	1.85%	0.58%	1.95%	0.87%	2.13%	0.77%	1.98%	0.53%
Annualized	0.31%	0.53%	0.31%	0.58%	0.32%	0.87%	0.35%	0.77%	0.33%	0.53%

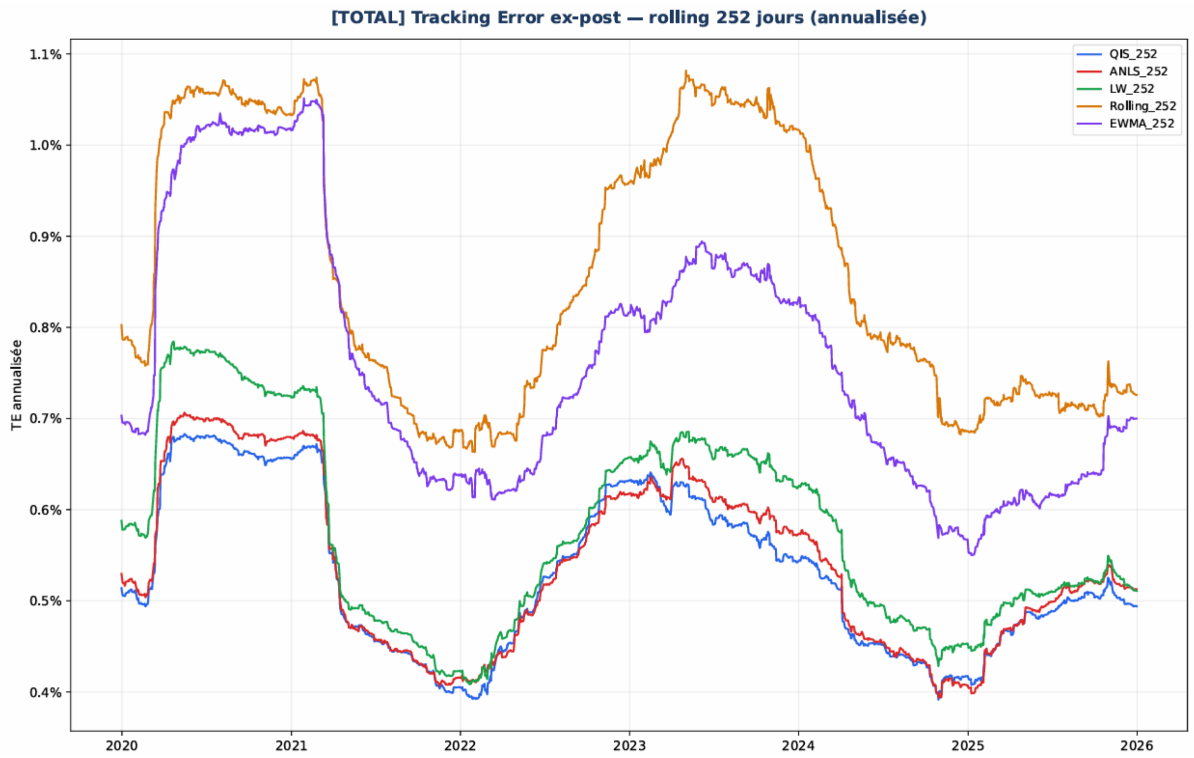

Figure 10: Ex-post tracking error (rolling 252 days) — M1US with exclusions (252-day window).

On the 252-day window, tracking error is high and very dispersed, reflecting an unfavorable p/n ratio in a restricted universe. Shrinkage models stand out clearly: QIS is the most stable, with levels between 0.40% and 0.66%, followed closely by ANLS, slightly more volatile in 2020 and 2022. LW is the least performant of the three shrinkage estimators, with systematically higher values, notably in 2020 (0.72%) and 2022 (0.66%).

Rolling and EWMA present significantly higher levels, regularly exceeding 0.70% and reaching over 1% in 2020. Stress years (2022–2023) accentuate these gaps: shrinkage models remain contained, while Rolling and EWMA see their tracking error surge. The chart confirms this dynamic, with shrinkage curves relatively parallel and marked peaks for Rolling and especially EWMA.

504-Day Window

Table 50: Annual results — M1US with exclusions (504-day window, period 2020–2025)

	ANLS 504d		LW 504d		Rolling 504d		EWMA 504d		QIS 504d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	1.87%	0.83%	1.16%	1.02%	1.62%	0.84%	1.07%	1.18%	1.07%	0.57%
2021	0.03%	0.47%	0.86%	0.16%	0.37%	0.40%	0.67%	0.54%	0.01%	0.41%
2022	-0.45%	0.62%	-0.63%	0.37%	-0.62%	0.41%	-0.55%	0.33%	-0.56%	0.62%
2023	1.43%	0.42%	0.66%	0.58%	1.01%	0.44%	0.66%	-0.20%	0.72%	0.51%
2024	2.30%	1.13%	2.15%	1.00%	2.23%	1.16%	2.10%	1.03%	0.85%	0.37%
2025	0.01%	-0.18%	-0.30%	-0.03%	-0.07%	-0.27%	-0.45%	-0.18%	0.00%	0.46%
Cumulative total	2.25%	0.63%	2.36%	0.63%	2.19%	0.54%	2.04%	0.78%	2.11%	0.50%
Annualized	0.37%	0.63%	0.39%	0.63%	0.36%	0.54%	0.34%	0.78%	0.35%	0.50%

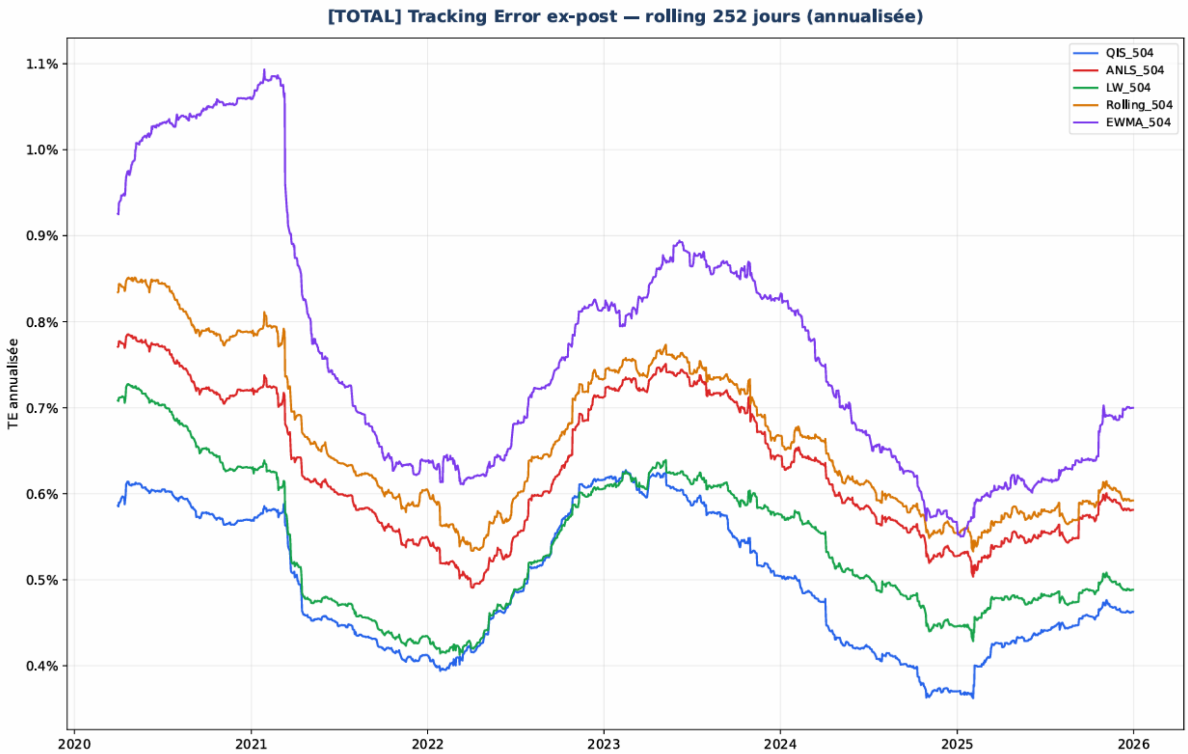

Figure 11: Ex-post tracking error (rolling 252 days) — M1US with exclusions (504-day window).

With a 504-day window, tracking error decreases for all models, reflecting the stabilizing effect of increasing observations. QIS remains the best- performing model, with levels between 0.37% and 0.62%. ANLS follows closely, but presents a punctual instability in 2024 where its tracking error exceeds 1%. LW sits in an intermediate position: globally stable, but more sensitive to tension episodes, notably in 2020 and 2024.

Rolling benefits from window lengthening, but remains systematically above the shrinkage models. EWMA remains the most volatile, with marked peaks in 2020 and 2024. The chart shows progressive convergence of shrinkage models, while Rolling and EWMA retain more erratic trajectories.

756-Day Window

Table 51: Annual results — M1US with exclusions (756-day window, period 2020–2025)

	ANLS 756d		LW 756d		Rolling 756d		EWMA 756d		QIS 756d	
Year	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE	Act. Rtn	TE
2020	1.19%	0.47%	0.68%	0.69%	0.70%	0.59%	1.19%	0.46%	0.70%	0.59%
2021	-0.42%	0.12%	0.49%	0.25%	-0.17%	0.40%	0.55%	0.64%	-0.17%	0.40%
2022	-0.55%	0.37%	0.65%	0.58%	-0.32%	0.58%	-0.33%	0.82%	-0.32%	0.58%
2023	1.00%	0.37%	0.63%	0.58%	0.34%	0.63%	-0.20%	0.83%	0.51%	0.51%
2024	2.40%	0.97%	0.47%	2.08%	0.98%	2.09%	1.03%	0.57%	0.88%	0.37%
2025	0.10%	0.11%	0.53%	0.21%	0.11%	0.53%	-0.18%	0.70%	0.04%	0.47%
Cumulative total	1.67%	0.57%	1.70%	0.57%	1.69%	0.58%	1.31%	0.71%	1.66%	0.49%
Annualized	0.29%	0.57%	0.29%	0.57%	0.29%	0.58%	0.23%	0.71%	0.29%	0.49%

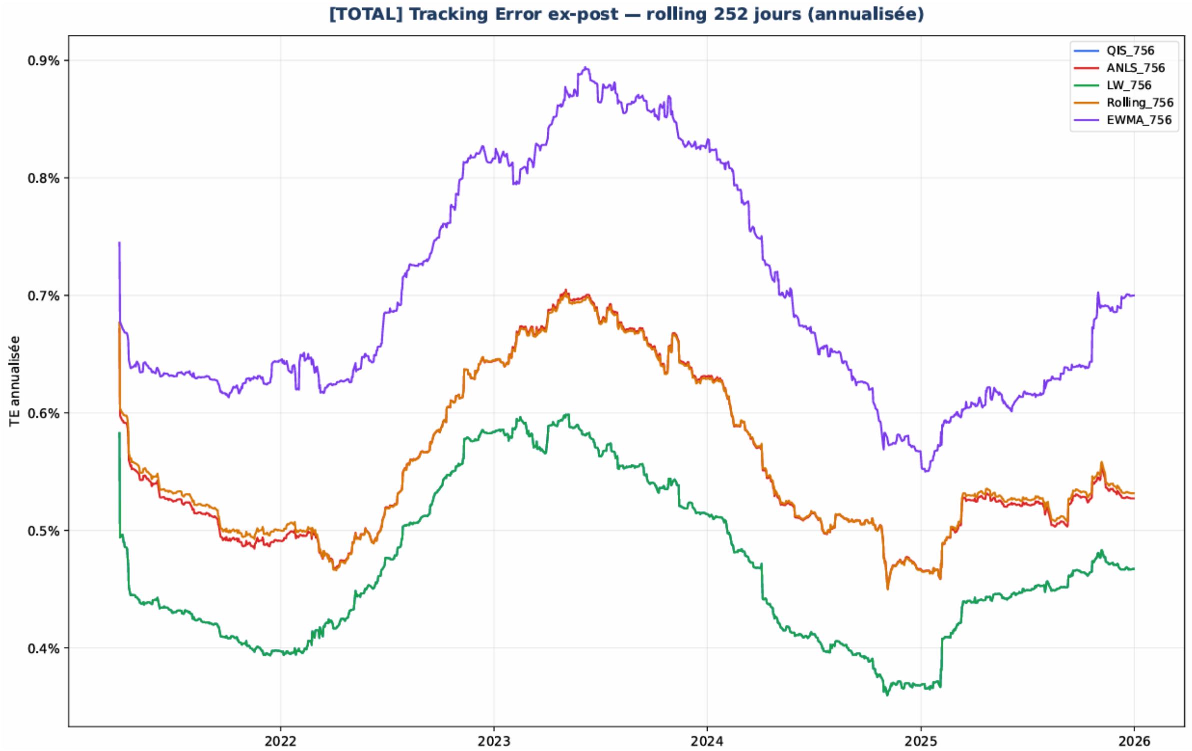

Figure 12: Ex-post tracking error (rolling 252 days) — M1US with exclusions (756-day window).

On the 756-day window, tracking error further decreases and trajectories converge more, but without reversing the hierarchy observed at shorter horizons. QIS remains the most robust model, with levels between 0.37% and 0.58%. ANLS follows closely, but presents increased volatility in 2024. LW remains slightly above, confirming its relative sensitivity to tension episodes.

Rolling approaches the shrinkage models, but remains penalized by empirical estimator noise. EWMA retains the highest levels, despite the overall decrease from window lengthening. The chart illustrates near-superposition of shrinkage and Rolling curves, while EWMA remains clearly above.